

Artificial Intelligence in Pediatric Radiology

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Objectives

- Describe the big players in radiology AI, both general and pediatric, since 2023
- Summarize what radiology AI does well and what remains unresolved

Methods

Journals, Preprints: PubMed, Embase, arXiv, medRxiv

Conferences: NeurIPS, ICLR, ICML, CVPR, MICCAI, MIDL

Datasets: Hand-curated list

Newsletters: The Imaging Wire, RSNA News, TLDR AI, TLDR Tech, Signify Research, ESR / ECR, Radiology Business

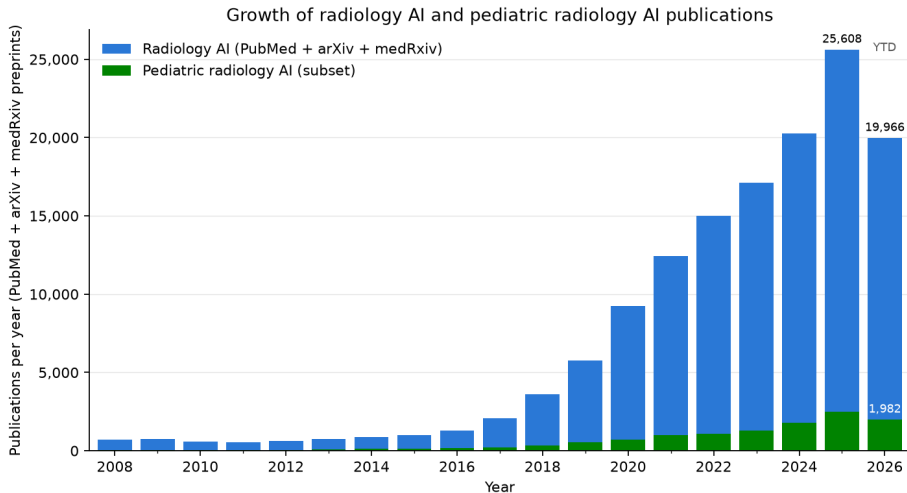
Commercial: FDA list of AI-enabled devices

Search strategy: radiology AND AI [AND pediatric]

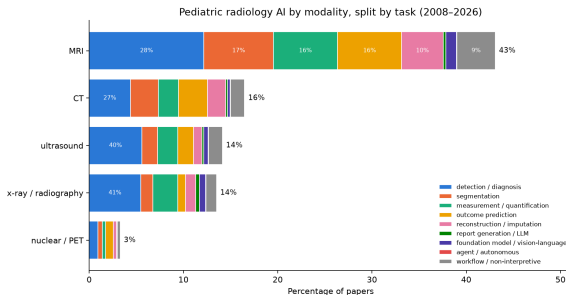
radiology: (radiology[tiab] OR radiological[tiab] OR radiograph*[tiab] OR "medical imaging"[tiab] OR "diagnostic imaging"[tiab] OR tomography[tiab] OR "magnetic resonance"[tiab] OR MRI[tiab] OR "MR imaging"[tiab] OR "MR images"[tiab] OR MRIs[tiab] OR "MR image"[tiab] OR "MR scan"[tiab] OR "MR scans"[tiab] OR "computed tomography"[tiab] OR CT[tiab] OR ultrasound[tiab] OR ultrasonograph*[tiab] OR sonograph*[tiab] OR echocardiograph*[tiab] OR mammograph*[tiab] OR "chest x-ray"[tiab] OR "x-ray"[tiab] OR fluoroscop*[tiab] OR scintigraph*[tiab] OR "positron emission"[tiab] OR PET[tiab] OR SPECT[tiab] OR angiograph*[tiab] OR neuroimaging[tiab] OR tractograph*[tiab] OR elastograph*[tiab] OR "fMRI"[tiab] OR "functional MRI"[tiab] OR "functional magnetic resonance"[tiab] OR connectome*[tiab] OR "diffusion tensor"[tiab] OR DTI[tiab] OR "bone age"[tiab] OR "skeletal age"[tiab] OR "skeletal maturity"[tiab] OR "barium enema"[tiab])

AI: ("artificial intelligence"[tiab] OR "machine learning"[tiab] OR "deep learning"[tiab] OR "convolutional neural network"[tiab] OR "convolutional neural networks"[tiab] OR "neural network"[tiab] OR "neural networks"[tiab] OR "computer-aided diagnosis"[tiab] OR "computer aided diagnosis"[tiab] OR "computer-aided detection"[tiab] OR radiomic*[tiab] OR "computer vision"[tiab] OR "large language model"[tiab] OR "large language models"[tiab] OR "foundation model"[tiab] OR "foundation models"[tiab] OR "transfer learning"[tiab] OR "vision transformer"[tiab] OR "self-supervised"[tiab] OR "random forest"[tiab] OR "support vector machine"[tiab] OR "gradient boosting"[tiab] OR "natural language processing"[tiab])

pediatric: (pediatric*[tiab] OR paediatric*[tiab] OR child*[tiab] OR infant*[tiab] OR neonat*[tiab] OR adolescen*[tiab] OR fetal[tiab] OR foetal[tiab] OR fetus*[tiab] OR foetus*[tiab] OR newborn*[tiab] OR preterm[tiab] OR "premature infant"[tiab] OR "premature infants"[tiab] OR "premature birth"[tiab] OR "premature neonate"[tiab] OR "premature neonates"[tiab] OR prenatal[tiab] OR antenatal[tiab] OR perinatal[tiab] OR youth[tiab] OR juvenile[tiab] OR "children's hospital"[tiab] OR schoolchild*[tiab] OR toddler*[tiab])

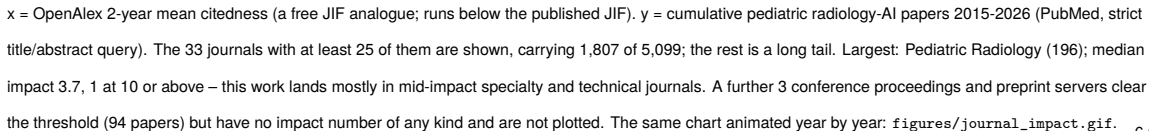


Pediatric radiology AI publications, stratified by modality and task

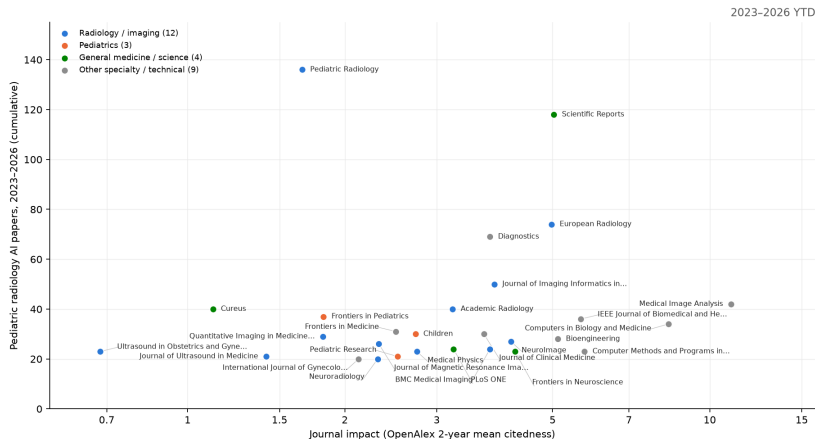


Same construction, restricted to the pediatric subset (10,737 papers incl. 596 preprints). Top pediatric modalities: MRI 41%, CT 16%, ultrasound 13%, x-ray / radiography 12%. Mammography is omitted: its 79 pediatric-query hits (of 10,737) are adult breast-imaging papers whose abstracts mention children.

- detection / diagnosis:** is a finding present now, and which one (label or box): fracture, pneumonia, tumor; includes triage and screening
- segmentation:** outline a structure or lesion (a mask)
- measurement / quantification:** a number from the image: bone age, Cobb angle, organ volume, fetal biometry
- outcome prediction:** a future risk, response, or survival estimate from the image (prognosis, radiomics signatures)
- reconstruction / imputation:** a better or missing image: lower dose, faster scans, denoising, synthetic CT/MR
- report generation / LLM:** text: draft, summarize, or extract from radiology reports
- foundation model / vision-language:** a large pretrained model reused across tasks (segment-anything, self-supervised)
- agent / autonomous:** multi-step actions taken without a human in the loop
- workflow / non-interpretive:** protocoling, scheduling, ordering, decision support, education



Which journals publish pediatric radiology AI, 2023–present



x = OpenAlex 2-year mean citedness (a free JIF analogue; runs below the published JIF). y = cumulative pediatric radiology-AI papers 2023-2026 (PubMed, strict title/abstract query). The 28 journals with at least 20 of them are shown, carrying 1,099 of 3,340; the rest is a long tail. Largest: Pediatric Radiology (136); median impact 3.2, 1 at 10 or above – this work lands mostly in mid-impact specialty and technical journals. medRxiv clears the threshold (31 papers) but has no impact number of any kind, so it is not plotted.

Cross-check: an independent 2025 scoping review of pediatric radiology AI

Kamran et al., *Pediatric Radiology* 2025 (doi 10.1007/s00247-025-06462-5): 789 original pediatric-focused articles hand-screened from four databases, 2005 to August 2024. Our PubMed counts are larger because they include any paper that mentions a pediatric term.

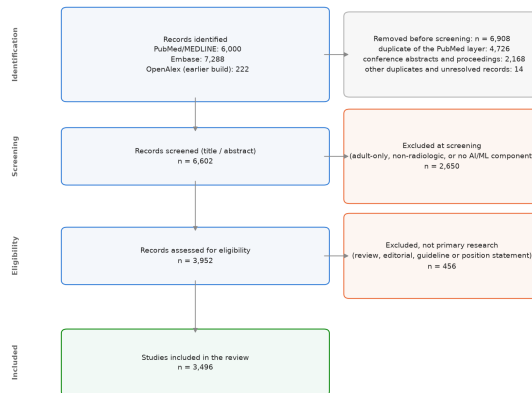
- **Modality:** radiography 38%, MRI 33%, ultrasound 14%, CT 11%, nuclear 2%. Our pediatric query ranks MRI first (MRI 41%, CT 16%, ultrasound 13%, x-ray / radiography 12%) because MRI-heavy neurodevelopment papers mention children.
- **Subspecialty:** musculoskeletal 33%, neuro 29%, chest 17%, body 14%, cardiac 6%.
- **Top applications:** bone age (88 articles), pneumonia (65), brain tumors (54), scoliosis (45), hip dysplasia (28), congenital heart disease (25), autism (18), epilepsy (14), ADHD (9), hydrocephalus (9).
- **Use of AI:** 91% image interpretation / diagnosis; 5.6% image quality; under 2% for acquisition, communication, protocoling, education, and policy combined.
- **Data:** 65% used a single local hospital dataset; 7% did not report the dataset; China 28%, USA 25%, Canada 7% of articles.
- **Growth:** 23 articles in 2018, 158 in 2022, 171 in 2023, consistent with our PubMed curve.
- **Recommendations:** multi-institutional, age-diverse pediatric datasets; CLAIM-style reporting with subgroup results and calibration; funding for the neglected non-interpretive areas (policy, education, implementation); children and families as stakeholders across the AI lifecycle.
- **What it adds to this deck:** the same three leading tasks (bone age, pneumonia, brain tumor) and the same gap (small local datasets, adult tools applied to children) reached by a manual review.

Reading the whole pediatric literature, not a sample of it

Counting publications says how much is being published; it does not say what was built, for which children, or how well it was tested. So every pediatric radiology-AI record from 2005 onward was retrieved from **PubMed/MEDLINE and Embase**, screened against prespecified criteria, and read into a structured row — model, modality, body region, age group, task, dataset size, data source, validation, availability — under one fixed schema.

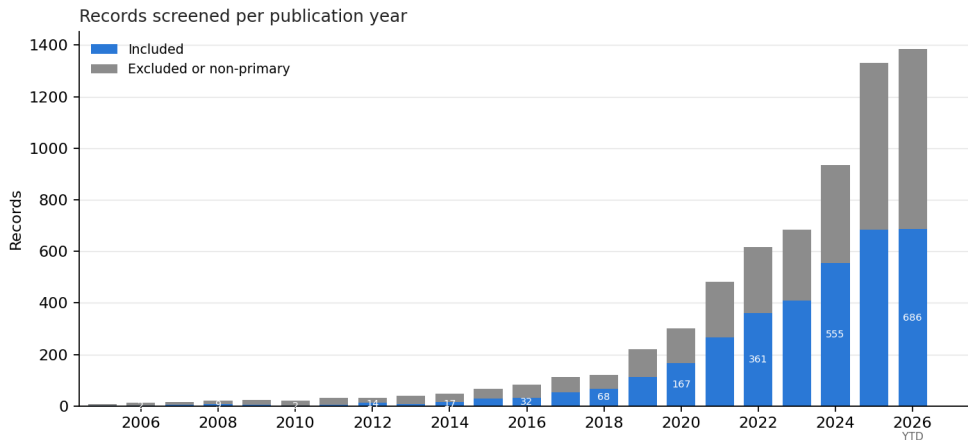
- 6,995 records retrieved, 393 conference proceedings set aside, 6,602 screened, 2,650 off-topic and 456 non-primary removed, 3,496 primary studies included (PubMed 3,122, Embase 365, OpenAlex 9).
- **Embase earns its place:** 365 of the included studies are not in PubMed at all. They have no PMID, so no NIH relative citation ratio; ranking them at all is why the impact column below is OpenAlex's field-weighted score.
- Conference proceedings are retrieved and reported but not screened: an abstract cannot support a judgement about study design, and it carries no DOI linking it to the later full paper.
- Screening and extraction were done by a constrained model under a fixed schema, with a blinded second-model re-screen of 150 records (94.7% agreement, $\kappa = 0.885$). Dual-human validation is outstanding.
- The same corpus is the paper database (`reports/04_paper_database.md`) and the systematic review manuscript, so the three never disagree.

Study selection



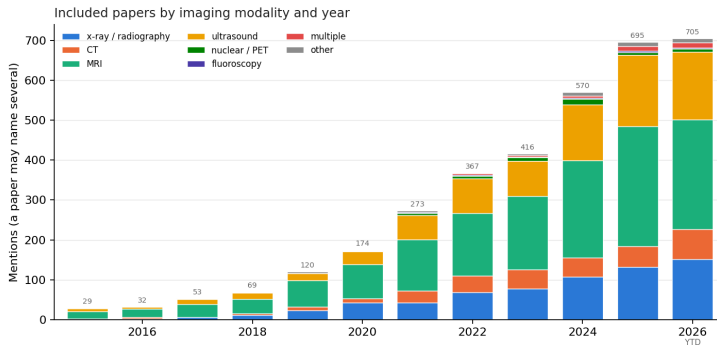
PubMed and Embase layers complete (2005-2026). Conference material is retrieved and reported but excluded by the publication-form criterion.

Included studies per publication year



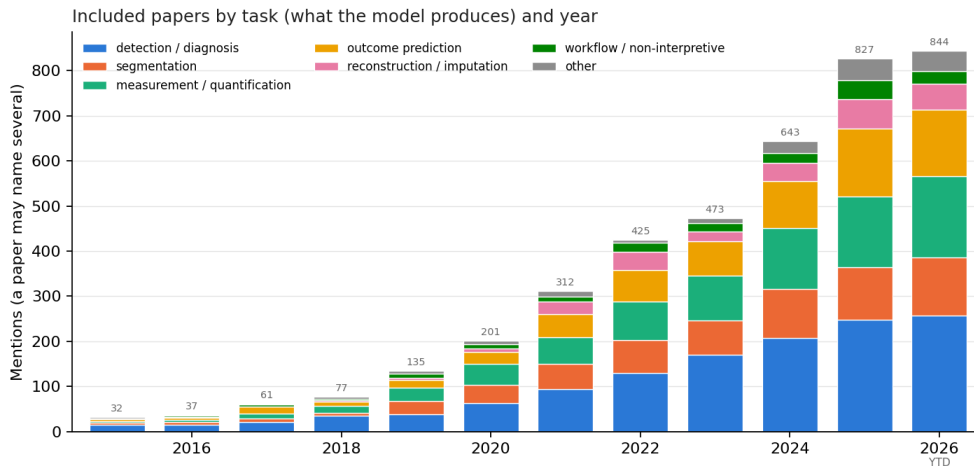
Blue = the 3,496 included primary studies; gray = records read and then excluded as off-topic or non-primary. 2025+ is year to date and still being indexed.

What the pediatric literature is made of

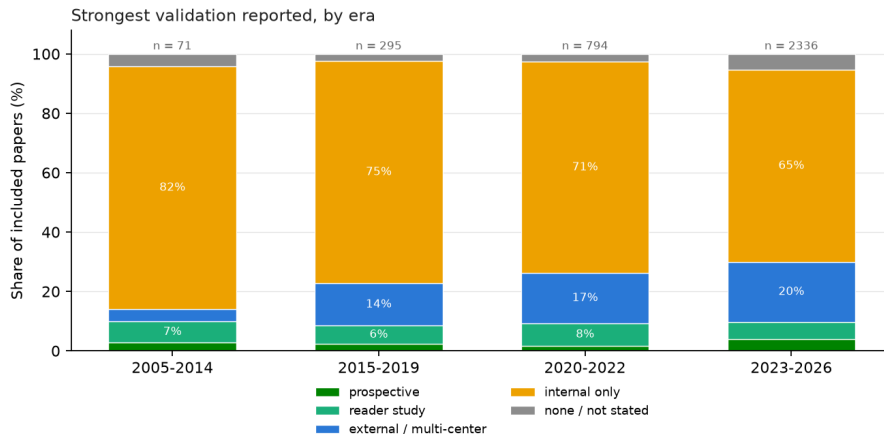


Modality: MRI 46%, ultrasound 24%, x-ray / radiography 19%, CT 9%. **Task:** detection / diagnosis 38%, measurement / quantification 24%, outcome prediction 20%, segmentation 19%. **Age group:** child 37%, adolescent 25%, fetal 22%, infant 15%. Categories are multilabel, so an axis can exceed 100%. MRI leads on the strength of the fetal and developmental-neuroscience literature: remove the MRI studies that use cognitive-neuroscience methods or report cognitive and behavioral outcomes and MRI falls level with ultrasound (30% each), which is why reviews built on narrower vocabulary rank radiography or ultrasound first.

What the models produce

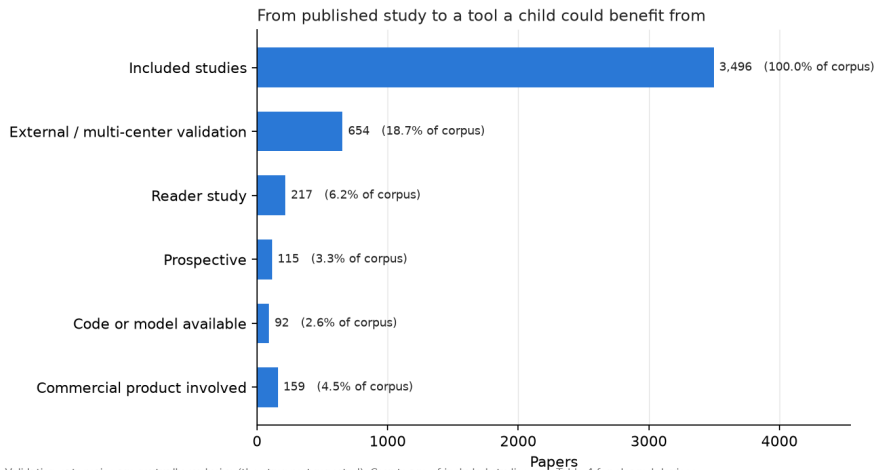


How well is it tested?



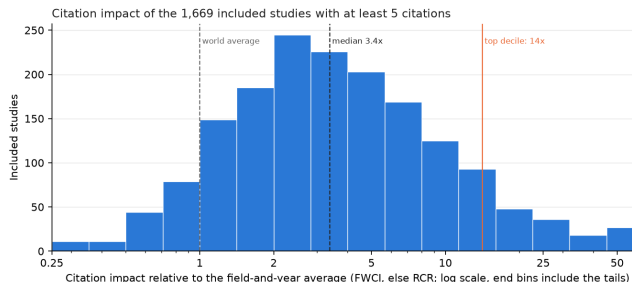
Of 3,496 included studies, external or multi-center validation was reported by 18.7%, a reader study by 6.2%, prospective evaluation by 3.3%, and a working code or model URL by 92 studies. External validation rose across eras; reader studies did not.

From published study to a tool a child benefits from



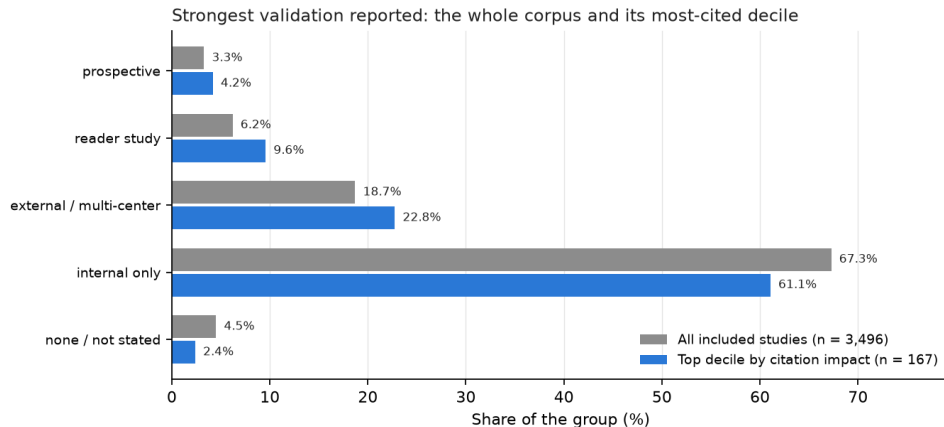
Validation categories are mutually exclusive (the strongest reported). Counts are of included studies; see Table 4 for cleared devices.

Which of these studies is the field actually reading?



- The median included study is cited 3.38 times the world average for its field and year: this is a well-cited literature, not a neglected one.
- 1,669 of 3,496 studies (47.7%) have enough citations to be ranked; the rest are too recent.
- The top decile is the 167 studies at 14x average and up (PubMed 158, Embase 6, OpenAlex 3).
- Selecting on citations does not select for rigor: external or multi-center validation is 22.8% in that decile against 18.7% across the corpus, and two thirds of it is still internal validation only.

Does citation impact pick out the better-validated work?



The most-cited decile is somewhat more likely to report external validation and a reader study, but the difference is small: citations track topic and audience more than evidence.

Highest-impact pediatric radiology AI studies, 2005–2026

Impact	Cites	Year	Study	Venue	What it does
329x	273	2018	Performance of a Deep-Learning Neural Network Model in Assessing Skeletal Maturity on Pediatric Hand Radiographs.	Radiology	skeletal maturity (bone age) assessment in the workup of endocrine and metabolic disorders
325x	219	2017	Fully Automated Deep Learning System for Bone Age Assessment.	Journal of digital imaging	skeletal maturity (bone age) assessment in endocrine and metabolic workup
293x	1,757	2018	Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning.	Cell	pediatric pneumonia on chest radiographs, and bacterial vs viral origin
221x	167	2017	Deep learning for automated skeletal bone age assessment in X-ray images.	Medical image analysis	skeletal maturity (bone age) assessment in endocrine, genetic and growth disorders
200x	321	2016	Automatic Segmentation of MR Brain Images With a Convolutional Neural Network.	IEEE transactions on medical imaging	automatic brain tissue segmentation for quantitative MRI analysis
93x	269	2021	CA-Net	IEEE transactions on medical imaging	segmenting the placenta and fetal brain on fetal MRI
83x	701	2019	Attention Gate (Attention U-Net)	Medical image analysis	identifying the correct standard views during fetal ultrasound screening
73x	112	2017	Computerized Bone Age Estimation Using Deep Learning Based Program: Evaluation of the Accuracy and Efficiency.	AJR. American journal of roentgenology	bone age estimation, and whether software assistance makes radiologists faster without making them worse
71x	78	2019	Artificial intelligence-assisted interpretation of bone age radiographs improves accuracy and decreases variability.	Skeletal radiology	whether AI assistance makes pediatric radiologists better at bone age, not whether AI alone beats them

Impact = citations relative to the average paper of the same field and year (OpenAlex FWCI; iCite RCR where OpenAlex has none), shown once a study has 5 citations. † Embase-only; * preprint.

Highest-impact pediatric radiology AI studies, 2023

Impact	Cites	Study	Venue	What it does
69x	106	iBEAT V2.0	Nature protocols	measuring cortical development in infants, whose brain MRI defeats adult-designed processing tools
37x	20	Bone Age Assessment Using Artificial Intelligence in Korean Pediatric Population: A Comparison of Deep-Learning Models Trained With Healthy Chronological and Greulich-Pyle Ages as Labels.	Korean journal of radiology	whether bone age in a given population should be predicted against the Greulich-Pyle atlas or learned directly from the chronological ages of healthy children in that population
34x	9	Bone age assessment based on deep neural networks with annotation-free cascaded critical bone region extraction.	Frontiers in artificial intelligence	automated bone age assessment from hand radiographs without manual region annotation
30x	39	FeTA	Medical image analysis	segmenting the developing fetal brain into tissues, the step every quantitative prenatal neurodevelopment study depends on
28x	20	Deep focus approach for accurate bone age estimation from lateral cephalogram.	Journal of dental sciences	estimating skeletal maturity from the lateral cephalogram a growing child already has, so that a separate hand-wrist radiograph is not required
27x	11	Bone Age Estimation and Prediction of Final Adult Height Using Deep Learning.	Yonsei medical journal	automatically estimating bone age and predicting final adult height in children using hand radiographs
26x	21	Detection of ASD Children through Deep-Learning Application of fMRI.	Children (Basel, Switzerland)	early detection of autism spectrum disorder in children, where prompt diagnosis enables targeted intervention
26x	14	High performance for bone age estimation with an artificial intelligence solution.	Diagnostic and interventional imaging	How does an AI solution for bone age assessment compare in accuracy to a general (non-pediatric-specialized) radiologist?
25x	49	Automated classification of common maternal fetal ultrasound planes using multi-layer perceptron with deep feature integration †	Biomedical Signal Processing and Control	automated identification of standard fetal ultrasound planes during obstetric examination

410 studies included from 2023; 289 have enough citations to be ranked. Impact = citations relative to the average paper of the same field and year (OpenAlex FWCI; iCite RCR where OpenAlex has none), shown once a study has 5 citations. † Embase-only; * preprint.

Highest-impact pediatric radiology AI studies, 2024

Impact	Cites	Study	Venue	What it does
29x	16	Automated bone age assessment in a German pediatric cohort: agreement between an artificial intelligence software and the manual Greulich and Pyle method.	European radiology	whether a bone age product can be swapped in for an expert reader, not merely correlate with one
29x	16	A Dempster-Shafer Approach to Trustworthy AI With Application to Fetal Brain MRI Segmentation.	IEEE transactions on pattern analysis and machine intelligence	detecting and correcting failures in automated fetal brain MRI segmentation so results are safe to use clinically
26x	12	HFSCCD: A Hybrid Neural Network for Fetal Standard Cardiac Cycle Detection in Ultrasound Videos.	IEEE journal of biomedical and health informatics	Automatically detecting standard cardiac cycle frames in fetal echocardiography to support congenital heart disease diagnosis
24x	5	Assessment of Bone Age Based on Hand Radiographs Using Regression-Based Multi-Modal Deep Learning.	Life (Basel, Switzerland)	estimating bone age from hand radiographs
24x	17	FetalBrainAwareNet	Computerized medical imaging and graphics	Synthesizing anatomically accurate fetal head standard-plane ultrasound images to augment training data for automated fetal head standard-plane classification.
24x	16	On the use of contrastive learning for standard-plane classification in fetal ultrasound imaging.	Computers in biology and medicine	automated identification of standard imaging planes in fetal ultrasound examinations
22x	12	Transformer based multi-modal MRI fusion for prediction of post-menstrual age and neonatal brain development analysis.	Medical image analysis	Estimating post-menstrual age and assessing brain maturity from neonatal brain MRI to help identify preterm-related developmental risk
21x	9	TW3 AI bone age assessment system	Quantitative imaging in medicine and surgery	validating an AI system for Tanner-Whitehouse 3 bone age assessment across multiple children's hospitals
20x	11	Artificial intelligence assisted common maternal fetal planes prediction from ultrasound images based on information fusion of customized convolutional neural networks.	Frontiers in medicine	automatically classifying standard maternal-fetal ultrasound planes/organs to aid interpretation

555 studies included from 2024; 264 have enough citations to be ranked.

Highest-impact pediatric radiology AI studies, 2025

Impact	Cites	Study	Venue	What it does
54x	10	ChatGPT-o4-mini-high, ChatGPT-4.5, Gemini 2.5 Pro	Academic radiology	whether a general-purpose chatbot can substitute for dedicated bone age software, which is expensive
52x	34	Prediction of mental health risk in adolescents.	Nature medicine	predicting future risk of general psychiatric illness in adolescents
32x	13	BoneXpert, PANDA, BoneView	European radiology	which of the bone age products approved for the European market a department should buy
29x	10	Whole examination AI estimation of fetal biometrics from 20-week ultrasound scans.	NPJ digital medicine	fetal biometry currently rests on a handful of frames a sonographer chose, and the choice is a source of error
29x	8	Uncovering ethical biases in publicly available fetal ultrasound datasets.	NPJ digital medicine	whether publicly available fetal ultrasound image datasets used to train deep-learning prenatal-diagnosis algorithms are demographically and clinically representative
28x	13	Scanner-based real-time three-dimensional brain + body slice-to-volume reconstruction for T2-weighted 0.55-T low-field fetal magnetic resonance imaging.	Pediatric radiology	motion-corrected 3D fetal MRI has stayed a research technique because reconstruction happens offline, after the patient has gone home
26x	21	Predicting pediatric patient rehabilitation outcomes after spinal deformity surgery with artificial intelligence.	Communications medicine	telling an adolescent before scoliosis surgery how they are likely to feel afterwards, which radiographic correction alone does not predict
24x	10	Three-dimensional markerless surface topography approach with convolutional neural networks for adolescent idiopathic scoliosis screening.	Scientific reports	screening adolescents for scoliosis without the unnecessary referrals and radiographs that current screening generates
23x	5	Advancing prenatal healthcare by explainable AI enhanced fetal ultrasound image segmentation using U-Net++ with attention mechanisms.	Scientific reports	automated, explainable segmentation of fetal ultrasound images to standardize evaluation of fetal development

685 studies included from 2025; 118 have enough citations to be ranked.

Highest-impact pediatric radiology AI studies, 2026 (year to date)

Impact	Cites	Study	Venue	What it does
46x	7	Infants have rich visual categories in ventrotemporal cortex at 2 months of age.	Nature neuroscience	characterizing the development of visual category representations in the infant brain using functional MRI, compared against a deep neural network model of vision
14x	6	Perceived Racism, Brain Development, and Internalizing and Externalizing Symptoms: Findings From the ABCD Study.	Journal of the American Academy of Child and Adolescent Psychiatry	assessing effects of perceived racial discrimination on longitudinal subcortical functional connectivity changes and internalizing/externalizing symptoms in adolescents
6x	5	Use of Artificial Intelligence-Based Software to Aid in the Identification of Ultrasound Findings Associated With Fetal Congenital Heart Defects.	Obstetrics and gynecology	evaluating whether AI-based software improves ob-gyn and maternal-fetal medicine specialists' identification of second-trimester fetal ultrasound findings suspicious for congenital heart defects
5x	5	Artificial Intelligence for the Detection of Fetal Ultrasound Findings Concerning for Major Congenital Heart Defects.	Obstetrics and gynecology	identifying second-trimester fetal ultrasound findings suspicious for major congenital heart defects using AI-based software, to improve antenatal CHD detection
4x	14	G-YOLOv11 †	Biomedical Signal Processing and Control	fracture detection on pediatric wrist radiographs
4x	8	Machine Learning Algorithms for Predicting Cobb Angle Beyond 25° in Female Adolescent Idiopathic Scoliosis Patients.	Spine	predicting which female adolescents with idiopathic scoliosis will progress to a Cobb angle beyond 25°, using minimal radiographic and questionnaire data collected at the first visit
3x	5	Motion-Echo	European heart journal	automating myocardial strain analysis from pediatric echocardiograms to standardize measurement and enable earlier detection of cardiac dysfunction
3x	8	Safe for kids? AI medical devices in radiology overlook paediatric suitability.	European radiology	whether the AI devices a European radiology department can buy say anything about being safe for children
3x	9	GPT-4.1	European archives of oto-rhino-laryngology	detecting and lateralizing pediatric airway foreign body aspiration from clinical data and chest radiographs

686 studies included from 2026; 12 have enough citations to be ranked. Few studies from this year have enough citations for a normalized value yet, so this table is thin by construction rather than because the year was quiet.

The biggest public pediatric radiology AI datasets

Dataset	Year	Modality	Size	Ages	Access
CBTN clinical MRI (Children's Brain Tumor Network)	2023	brain / spine tumor MRI	23,101 exams (1,526 patients); tumor masks for 370	pediatric	application (NCI CCDI)
GRAZPEDWRI-DX	2022	wrist radiographs (trauma)	20,327 images (6,091 patients)	0.2–19 y	open
RSNA Pediatric Bone Age Challenge	2017	hand radiographs	14,236 images (bone age + sex)	mean 10.6 y	open (RSNA terms)
FETAL_PLANES_DB	2020	fetal ultrasound (standard planes)	12,400 images (1,792 pregnancies)	fetal	open
ABCD Study	2018–	brain MRI, longitudinal	11,878 participants, up to 7 follow-ups	9–10 y at baseline	application (NBDC)
PediCXR / VinDr-PCXR	2023	chest radiographs	9,125 studies; 36 findings, 15 diagnoses	< 10 y	credentialed (PhysioNet)
Guangzhou pediatric chest X-ray (Kermany)	2018	chest radiographs	5,856 images (normal / bacterial / viral pneumonia)	1–5 y	open
ABIDE I + II (autism)	2012 / 2016	brain MRI (structural, resting fMRI)	2,156 subjects	5–64 y (mostly children)	open
Regensburg Pediatric Appendicitis	2023	abdominal ultrasound + labs	1,709 images (579 patients)	0–18 y	open
dHCP (Developing Human Connectome Project)	2022–24	neonatal + fetal brain MRI	886 neonatal + 297 fetal datasets, with segmentations	20–45 weeks	open (NDA)
Pediatric-CT-SEG (TCIA)	2022	chest / abdomen / pelvis CT	718 series (359 patients); 29 organ contours	5 days – 16 y	open
BraTS-PEDs (pediatric brain tumor challenge)	2023–25	brain tumor MRI	464 patients (2024 challenge); 1,649 series on TCIA	pediatric	registration / TCIA
FeTA (fetal brain tissue annotation)	2021–24	fetal brain MRI	280 volumes (120 train + 160 test); 7 tissue labels	18–35 weeks	registration

Sizes verified against the primary paper or hosting page (2026-09). Only two pediatric radiograph sets exceed 10,000 images; the largest public pediatric-only CT set has 359 patients; no RSNA pediatric challenge has run since bone age in 2017. Adult benchmarks (CheXpert 224k, MIMIC-CXR 377k) exclude children.

Most-cited radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
14,326	1023.0	A survey on deep learning in medical image analysis (2017)	Medical Image Anal.	<i>survey</i> : What can deep learning do across medical imaging? (the field's reference survey)
5,658	446.6	Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation (2021)	ECCV Workshops	<i>segmentation</i>
2,990	340.4	Deep Learning in Medical Image Analysis (2017)	Annual Review of Biomedical Engineering	<i>survey</i> : How does deep learning apply to medical image analysis? (review)
2,553	290.0	Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images (2022)	BrainLes@MICCAI	<i>segmentation</i>
2,769	272.4	COVID-Net: a tailored deep convolutional neural network design for detection of COVID-19 cases from chest X-ray images (2020)	Scientific Reports	<i>classification</i> : Can a CNN detect COVID-19 on chest X-ray? (COVID-Net)
3,634	236.9	ChestX-Ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases (2017)	Computer Vision and Pattern Recognition	<i>classification</i>
10,044	226.3	nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation (2020)	Nature Methods	<i>segmentation</i> : Can one self-configuring method segment any organ or lesion without hand tuning? (nnU-Net)
4,100	170.2	Convolutional neural networks: an overview and application in radiology (2018)	Insights into Imaging	<i>tutorial</i> : What is a convolutional neural network, explained for radiologists?
5,587	158.4	Computational Radiomics System to Decode the Radiographic Phenotype (2017)	Cancer Research	<i>radiomics</i> : Can standardized image features (radiomics) be extracted reproducibly? (PyRadiomics)
3,322	149.0	Artificial intelligence in radiology (2018)	Nature Reviews. Cancer	<i>review</i> : How will AI change radiology practice? (Hosny et al.)

Rows are the most-cited papers of the period, ordered by FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average), so a recent paper can be compared with an older one. Preprints (arXiv, medRxiv) are included; the venue column names the server.

Most-cited radiology AI papers, 2023

Cites	FWCI	Paper	Venue	What it answers
1,608	615.8	Segment anything in medical images	Nature Communications	<i>foundation</i> : Can a prompt-based 'segment anything' model work on medical images? (MedSAM)
2,099	243.8	Foundation models for generalist medical artificial intelligence	Nature	<i>foundation model</i> : What would a single generalist medical AI model that handles images, text and records look like? (perspective)
702	118.5	Redefining Radiology: A Review of Artificial Intelligence Integration in Medical Imaging	Diagnostics	<i>survey / review</i> : Where does AI fit in radiology workflow, from acquisition to reporting? (review)
509	75.2	Medical image analysis using deep learning algorithms	Frontiers in Public Health	<i>other</i>
562	74.5	A Study of CNN and Transfer Learning in Medical Imaging: Advantages, Challenges, Future Scope	Sustainability	<i>other</i>
861	69.6	Segment Anything Model for medical image analysis: an experimental study	Medical Image Anal.	<i>segmentation</i> : How well does the unmodified Segment Anything Model work on medical images? (it needs prompts and adaptation)
573	55.8	MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning techniques	BMC Medical Informatics and Decision Making	<i>oncology</i>
460	44.5	Brain Tumor Detection Based on Deep Learning Approaches and Magnetic Resonance Imaging	Cancers	<i>oncology</i>
391	42.2	SAM-Adapter: Adapting Segment Anything in Underperformed Scenes	2023 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)	<i>segmentation</i>
442	37.4	A generalist vision-language foundation model for diverse biomedical tasks	Nature Medicine	<i>other</i>

Rows are the most-cited papers of the period, ordered by FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average), so a recent paper can be compared with an older one. Preprints (arXiv, medRxiv) are included; the venue column names the server. 1 of the 10 rows are preprints.

Most-cited radiology AI papers, 2024

Cites	FWCI	Paper	Venue	What it answers
1,074	225.2	VM-UNet: Vision Mamba UNet for Medical Image Segmentation	ACM Transactions on Multimedia Computing, Communications, and Applications (TOMCCAP)	<i>segmentation</i> : Does a state-space (Mamba) backbone beat convolutions and transformers for medical segmentation?
242	110.3	Generalist foundation models from a multimodal dataset for 3D computed tomography	Nature Biomedical Engineering	<i>other</i>
282	74.6	Artificial intelligence in neuro-oncology: advances and challenges in brain tumor diagnosis, prognosis, and precision treatment	npj Precision Oncology	<i>oncology</i>
279	72.6	METHodological RadiomIcs Score (METRICS): a quality scoring tool for radiomics research endorsed by EuSoMI	Insights into Imaging	<i>radiomics</i> : How should the quality of a radiomics study be scored? (METRICS checklist)
349	47.6	Deep learning for medical image segmentation: State-of-the-art advancements and challenges	Informatics in Medicine Unlocked	<i>segmentation</i>
317	43.4	Advances in Medical Image Segmentation: A Comprehensive Review of Traditional, Deep Learning and Hybrid Approaches	Bioengineering	<i>segmentation</i>
238	37.0	Employing deep learning and transfer learning for accurate brain tumor detection	Scientific Reports	<i>oncology</i>
242	36.0	Deep semi-supervised learning for medical image segmentation: A review	Expert systems with applications	<i>segmentation</i>
241	34.5	A review of deep learning-based information fusion techniques for multimodal medical image classification	Comput. Biol. Medicine	<i>survey / review</i>
298	31.8	Vision-language models for medical report generation and visual question answering: a review	Frontiers Artif. Intell.	<i>report generation / LLM</i> : What can vision-language models do for report generation and visual question answering? (review)

Most-cited radiology AI papers, 2025

Cites	FWCI	Paper	Venue	What it answers
258	225.8	Deep Convolutional Neural Networks in Medical Image Analysis: A Review	Inf.	<i>survey / review</i>
145	158.8	Multi-model assurance analysis showing large language models are highly vulnerable to adversarial hallucination attacks during clinical decision support	Communications Medicine	<i>LLM / reports</i>
122	108.2	Multimodal generative AI for medical image interpretation	Nature	<i>other</i>
313	94.9	Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data	Nature Communications	<i>foundation model</i> : Can a radiology foundation model trained on web-scale 2D and 3D scans generalize across modalities and tasks? (RadFM)
173	87.2	Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology	Nature Cancer	<i>agent / autonomous</i> : Can an autonomous LLM agent that calls imaging and other tools support oncology decisions? (validation study)
208	65.9	A review of deep learning for brain tumor analysis in MRI	npj Precision Oncology	<i>survey / review</i>
150	65.8	Med-R1: Reinforcement Learning for Generalizable Medical Reasoning in Vision-Language Models	IEEE Transactions on Medical Imaging	<i>LLM / reports</i>
233	64.6	Artificial intelligence in healthcare and medicine: clinical applications, therapeutic advances, and future perspectives	European Journal of Medical Research	<i>other</i>
151	54.0	Medical Image Segmentation: A Comprehensive Review of Deep Learning-Based Methods	Tomography	<i>segmentation</i>
118	39.5	Advanced Brain Tumor Classification in MR Images Using Transfer Learning and Pre-Trained Deep CNN Models	Cancers	<i>oncology</i>

1 of the 10 rows are preprints.

Most-cited radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
48	102.6	A generalizable foundation model for analysis of human brain MRI	Nature Neuroscience	<i>other</i>
16	90.5	Reporting checklist for foundation and large language models in medical research (REFINE): an international consensus guideline.	Diagnostic and Interventional Radiology	<i>LLM / reports</i>
15	76.1	Scaling medical AI across clinical contexts	Nature Medicine	<i>foundation model</i> : How can medical AI be scaled across hospitals, populations and tasks without breaking? (perspective)
27	67.3	Applications and limitations of large language models to integrate medical context: a comprehensive review	Iran Journal of Computer Science	<i>survey / review</i>
18	62.7	A hybrid deep learning framework based on VGG19 and U-Net for accurate brain tumor segmentation in MRI images	Magnetic Resonance Letters	<i>segmentation</i>
37	58.8	Explainable artificial intelligence (XAI) in medical imaging: a systematic review of techniques, applications, and challenges	BMC Medical Imaging	<i>survey / review</i> : Which explainability methods are used in medical imaging AI and do they help? (review)
12	45.3	Explainable AI-driven MRI-based brain tumor classification: a novel deep learning approach	Frontiers Artif. Intell.	<i>oncology</i>
21	44.2	Agentic AI in Radiology: Evolution from Large Language Models to Future Clinical Integration.	Radiology: Artificial Intelligence	<i>agent / autonomous</i> : What are agentic AI systems in radiology and what stands between them and clinical use? (review)
17	28.4	Leveraging multi-modal foundation models for analysing spatial multi-omic and histopathology data	Nature Biomedical Engineering	<i>other</i>
19	21.4	BIBSNet: A deep learning baby image brain segmentation network for MRI scans	Developmental Cognitive Neuroscience	<i>segmentation</i> : Can one network segment infant brain MRI across the first years of life? (BIBSNet)

Citations for the current year are still accumulating; the FWCI column is the more stable signal here. 1 of the 10 rows are preprints.

Most-cited pediatric radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
426	328.6	Performance of a Deep-Learning Neural Network Model in Assessing Skeletal Maturity on Pediatric Hand Radiographs. (2017)	Radiology	<i>bone age</i> : Can a CNN assess skeletal maturity as accurately as radiologists?
431	321.5	Fully Automated Deep Learning System for Bone Age Assessment (2017)	Journal of digital imaging	<i>bone age</i> : Fully automated bone age from hand radiographs
403	221.2	Deep learning for automated skeletal bone age assessment in X-ray images (2017)	Medical Image Anal.	<i>bone age</i> : Deep learning for automated bone age (BoNet)
143	73.0	Computerized Bone Age Estimation Using Deep Learning Based Program: Evaluation of the Accuracy and Efficiency. (2017)	AJR. American journal of roentgenology	<i>bone age</i> : Is a commercial deep-learning bone-age program accurate and faster in practice?
191	57.8	A Review on Deep-Learning Algorithms for Fetal Ultrasound-Image Analysis (2022)	Medical Image Anal.	<i>survey / review</i>
352	57.4	Standard Plane Localization in Fetal Ultrasound via Domain Transferred Deep Neural Networks (2015)	IEEE journal of biomedical and health informatics	<i>fetal MRI</i>
194	57.0	Automatic Fetal Ultrasound Standard Plane Recognition Based on Deep Learning and IIoT (2021)	IEEE Transactions on Industrial Informatics	<i>fetal MRI</i>
233	30.3	Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes (2020)	Scientific Reports	<i>fetal MRI</i>
458	27.0	The RSNA Pediatric Bone Age Machine Learning Challenge. (2019)	Radiology	<i>bone age</i> : How well can crowdsourced AI estimate bone age? (RSNA challenge, 260 teams)
253	25.8	An ensemble of neural networks provides expert-level prenatal detection of complex congenital heart disease (2021)	Nature Medicine	<i>detection</i>

A different corpus from the systematic-review tables earlier in the deck: this is a citation search of Semantic Scholar over radiology AI with a pediatric title filter, so it includes conference papers and is not screened for eligibility. The review tables rank the screened corpus. Both are shown because neither contains the other.

Most-cited pediatric radiology AI papers, 2023

Cites	FWCI	Paper	Venue	What it answers
32	36.6	Bone Age Assessment Using Artificial Intelligence in Korean Pediatric Population: A Comparison of Deep-Learning Models Trained With Healthy Chronological and Greulich-Pyle Ages as Labels	Korean Journal of Radiology	<i>bone age</i>
62	33.6	Standard fetal ultrasound plane classification based on stacked ensemble of deep learning models	Expert systems with applications	<i>fetal ultrasound</i> : Can an ensemble of networks classify the standard fetal ultrasound planes?
27	33.5	Bone age assessment based on deep neural networks with annotation-free cascaded critical bone region extraction	Frontiers in Artificial Intelligence	<i>bone age</i>
43	23.8	Review on deep learning fetal brain segmentation from Magnetic Resonance images	Artif. Intell. Medicine	<i>segmentation</i>
37	20.2	Transfer learning for accurate fetal organ classification from ultrasound images: a potential tool for maternal healthcare providers	Scientific Reports	<i>fetal MRI</i>
34	16.6	Deep Learning-Based Multiclass Brain Tissue Segmentation in Fetal MRIs	Italian National Conference on Sensors	<i>segmentation</i>
43	9.7	Deeplasia: deep learning for bone age assessment validated on skeletal dysplasias	Pediatric Radiology	<i>bone age</i>
30	7.8	Deep learning for estimation of fetal weight throughout the pregnancy from fetal abdominal ultrasound.	American Journal of Obstetrics & Gynecology MFM	<i>fetal MRI</i>
38	7.3	Deep learning model for prenatal congenital heart disease (CHD) screening can be applied to retrospective imaging from the community setting, outperforming initial clinical detection in a well-annotated cohort	Ultrasound in Obstetrics and Gynecology	<i>fetal ultrasound</i> : Does a prenatal congenital heart disease screening model work on retrospective real-world scans?
31	5.8	MRI-Based End-To-End Pediatric Low-Grade Glioma Segmentation and Classification	Canadian Association of Radiologists journal = Journal l'Association canadienne des radiologistes	<i>segmentation</i>

Rows are the most-cited papers of the period, ordered by FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average), so a recent paper can be compared with an older one. Preprints (arXiv, medRxiv) are included; the venue column names the server. 1 of the 10 rows are preprints.

Most-cited pediatric radiology AI papers, 2024

Cites	FWCI	Paper	Venue	What it answers
17	34.8	Intrapartum Ultrasound Image Segmentation of Pubic Symphysis and Fetal Head Using Dual Student-Teacher Framework with CNN-VIT Collaborative Learning	International Conference on Medical Image Computing and Computer-Assisted Intervention	<i>segmentation</i>
30	23.8	FetalBrainAwareNet: Bridging GANs with anatomical insight for fetal ultrasound brain plane synthesis	Comput. Medical Imaging Graph.	<i>fetal MRI</i>
20	19.5	Artificial intelligence assisted common maternal fetal planes prediction from ultrasound images based on information fusion of customized convolutional neural networks	Frontiers in Medicine	<i>fetal MRI</i>
18	15.9	Deep learning denoising reconstruction for improved image quality in fetal cardiac cine MRI	Frontiers in Cardiovascular Medicine	<i>reconstruction</i> : Does deep-learning denoising improve fetal cardiac cine MRI?
30	15.1	RTSeg-net: A lightweight network for real-time segmentation of fetal head and pubic symphysis from intrapartum ultrasound images	Comput. Biol. Medicine	<i>segmentation</i>
16	13.4	FetSAM: Advanced Segmentation Techniques for Fetal Head Biometrics in Ultrasound Imagery	IEEE Open Journal of Engineering in Medicine and Biology	<i>segmentation</i>
23	7.0	Deep learning-based automated lesion segmentation on pediatric focal cortical dysplasia II preoperative MRI: a reliable approach	Insights into Imaging	<i>epilepsy</i> : Can focal cortical dysplasia lesions be found automatically on preoperative pediatric MRI?
18	5.3	Integration of a Deep Convolutional Neural Network With Adaptive Channel Weight Technique for Automated Identification of Standard Fetal Biometry Planes	IEEE Transactions on Instrumentation and Measurement	<i>fetal MRI</i>
18	5.2	Explainable AI in Deep Learning Based Classification of Fetal Ultrasound Image Planes	Procedia Computer Science	<i>fetal MRI</i>
19	4.8	A deep learning framework for identifying and segmenting three vessels in fetal heart ultrasound images	BioMedical Engineering OnLine	<i>segmentation</i>

Most-cited pediatric radiology AI papers, 2025

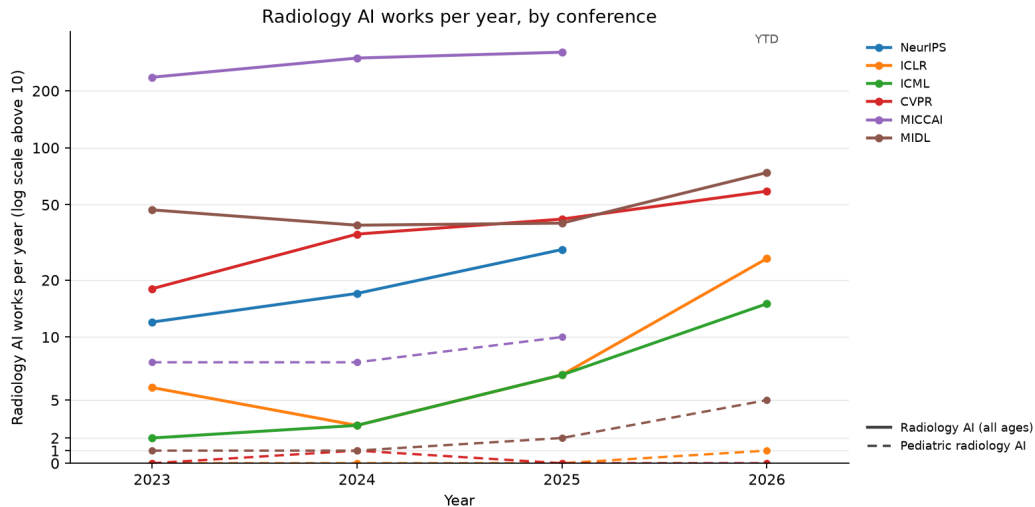
Cites	FWCI	Paper	Venue	What it answers
16	171.3	A comprehensive review of artificial intelligence - based algorithm towards fetal facial anomalies detection (2013–2024)	Artificial Intelligence Review	<i>survey / review</i>
18	85.1	M-CNN-RF: A hybrid deep learning model for accurate pediatric skeletal age estimation using hand bone radiographs	Alexandria Engineering Journal	<i>other</i>
20	77.0	Bone Age Assessment Using Various Medical Imaging Techniques Enhanced by Artificial Intelligence	Diagnostics	<i>bone age</i>
18	24.9	Advancing prenatal healthcare by explainable AI enhanced fetal ultrasound image segmentation using U-Net++ with attention mechanisms	Scientific Reports	<i>segmentation</i>
12	21.8	FetalMovNet: A Novel Deep Learning Model Based on Attention Mechanism for Fetal Movement Classification in US	IEEE Access	<i>fetal MRI</i>
11	14.5	Effectiveness and clinical impact of using deep learning for first-trimester fetal ultrasound image quality auditing	BMC Pregnancy and Childbirth	<i>fetal MRI</i>
12	12.4	Association between deep learning radiomics based on placental MRI and preeclampsia with fetal growth restriction: A multicenter study.	European Journal of Radiology	<i>radiomics</i>
16	8.7	Prenatal detection of congenital heart defects using the deep learning-based image and video analysis: protocol for Clinical Artificial Intelligence in Fetal Echocardiography (CAIFE), an international multicentre multidisciplinary study	BMJ Open	<i>fetal ultrasound: Can deep learning on prenatal ultrasound video detect congenital heart defects?</i>
11	8.6	Multi-view deep learning framework for the detection of chest X-rays compatible with pediatric pulmonary tuberculosis	Nature Communications	<i>detection</i>
11	7.0	Evaluation of Image Quality and Scan Time Efficiency in Accelerated 3D T1-Weighted Pediatric Brain MRI Using Deep Learning-Based Reconstruction	Korean Journal of Radiology	<i>reconstruction</i>

Most-cited pediatric radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
11	34.3	Artificial intelligence and radiomics for pediatric brain tumor classification and molecular characterization: a systematic review	Neuroradiology	<i>oncology</i> : Can AI and radiomics classify pediatric brain tumors and their molecular subtypes? (review)
5	27.9	Deep learning assessment of fetal brain maturation on 3D ultrasound volumes in early-onset fetal growth restriction	Ultrasound in Obstetrics and Gynecology	<i>fetal ultrasound</i> : Can deep learning grade fetal brain maturation on 3D ultrasound in growth restriction?
3	17.6	Foundation models in radiology: a primer for pediatric radiologists	Pediatric Radiology	<i>foundation model</i> : What should a pediatric radiologist know about foundation models? (primer)
2	17.6	Deep Learning-Based Multi-Class Pediatric Wrist Fracture Subtype Classification: A Pilot Study Comparing Convolutional Neural Network Architectures	Journal of Imaging	<i>fracture</i>
3	17.3	Current applications and challenges of artificial intelligence applied to diagnostics in pediatric musculoskeletal imaging	Pediatric Radiology	<i>classification</i>
4	13.6	Data mining in pediatric radiology in the era of artificial intelligence.	Pediatric Radiology	<i>other</i>
2	13.2	Acquisition time/dose reduction in pediatric PET imaging using patch-based deep learning	EJNMMI Physics	<i>reconstruction</i> : Can pediatric PET use one fifth of the counts with patch-based deep-learning denoising?
2	12.6	Optimizing pediatric chest CT: superior image quality and lower radiation dose with deep learning reconstruction.	European Journal of Radiology	<i>reconstruction</i> : Can deep-learning reconstruction lower pediatric chest CT dose while improving image quality?
2	12.2	Enhancing efficiency in pediatric brain tumor segmentation using a pathologically diverse single-center clinical dataset	Neuro-Oncology Advances	<i>segmentation</i>
3	11.7	Automated bone age assessment in rare pediatric growth disorders: a comparative study using Deeplasia	Frontiers in Endocrinology	<i>bone age</i> : Does an open bone-age model hold up in rare growth disorders against several experts? (Deeplasia)

Citations for the current year are still accumulating; the FWCI column is the more stable signal here.

Radiology AI works per year, by conference



NeurIPS: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	24	–	3D-RAD: A Comprehensive 3D Radiology Med-VQA Dataset with Multi-Temporal Analysis and Diverse Diagnostic Tasks
2025	20	–	Better Tokens for Better 3D: Advancing Vision-Language Modeling in 3D Medical Imaging
2024	47	2.0	Touchstone Benchmark: Are We on the Right Way for Evaluating AI Algorithms for Medical Segmentation?
2024	26	0.9	Eye-gaze Guided Multi-modal Alignment for Medical Representation Learning
2024	25	1.2	Enhancing vision-language models for medical imaging: bridging the 3D gap with innovative slice selection
2024	20	–	DiffusionBlend: Learning 3D Image Prior through Position-aware Diffusion Score Blending for 3D Computed Tomography Reconstruction
2023	144	–	SegVol: Universal and Interactive Volumetric Medical Image Segmentation
2023	104	–	LVM-Med: Learning Large-Scale Self-Supervised Vision Models for Medical Imaging via Second-order Graph Matching
2023	72	–	EHRXQA: A Multi-Modal Question Answering Dataset for Electronic Health Records with Chest X-ray Images
2023	66	–	SwiFT: Swin 4D fMRI Transformer
2023	64	–	NeuroGraph: Benchmarks for Graph Machine Learning in Brain Connectomics
2023	19	–	Knowledge-based in silico models and dataset for the comparative evaluation of mammography AI for a range of breast characteristics, lesion conspicuities and doses

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of the 45 works indexed there, newest year first.

Year	Cites	FWCI	Paper
2025	70	–	Large-scale and Fine-grained Vision-language Pre-training for Enhanced CT Image Understanding
2025	13	–	SimXRD-4M: Big Simulated X-ray Diffraction Data and Crystal Symmetry Classification Benchmark
2025	9	–	Self-Supervised Diffusion MRI Denoising via Iterative and Stable Refinement
2025	9	–	Score-based Self-supervised MRI Denoising
2024	143	–	MMed-RAG: Versatile Multimodal RAG System for Medical Vision Language Models
2024	31	–	MediConfusion: Can you trust your AI radiologist? Probing the reliability of multimodal medical foundation models
2024	19	–	The Effect of Intrinsic Dataset Properties on Generalization: Unraveling Learning Differences Between Natural and Medical Images
2024	16	–	LeFusion: Controllable Pathology Synthesis via Lesion-Focused Diffusion Models
2023	107	–	Advancing Radiograph Representation Learning with Masked Record Modeling
2023	84	–	DDM2: Self-Supervised Diffusion MRI Denoising with Generative Diffusion Models
2023	40	–	AE-FLOW: Autoencoders with Normalizing Flows for Medical Images Anomaly Detection
2023	18	–	Improved Training of Physics-Informed Neural Networks Using Energy-Based Priors: a Study on Electrical Impedance Tomography
2026	–	–	Fetal-Gauge: A Benchmark For Assessing Vision-Language Models In Fetal Ultrasound ★

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 13 most-cited of the 17 works indexed there, newest year first.

ICML: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	21	–	CLIMB: Data Foundations for Large Scale Multimodal Clinical Foundation Models
2025	17	–	MindLLM: A Subject-Agnostic and Versatile Model for fMRI-to-Text Decoding
2025	11	–	Raptor: Scalable Train-Free Embeddings for 3D Medical Volumes Leveraging Pretrained 2D Foundation Models
2025	5	–	Efficient Noise Calculation in Deep Learning-based MRI Reconstructions
2025	2	–	LangDAug: Langevin Data Augmentation for Multi-Source Domain Generalization in Medical Image Segmentation
2025	0	–	Staged and Physics-Grounded Learning Framework with Hyperintensity Prior for Pre-Contrast MRI Synthesis
2024	14	–	Unsupervised Domain Adaptation for Anatomical Structure Detection in Ultrasound Images
2024	10	–	Adaptive Sampling of k-Space in Magnetic Resonance for Rapid Pathology Prediction
2023	17	–	ACAT: Adversarial Counterfactual Attention for Classification and Detection in Medical Imaging
2023	16	–	Robustness of Deep Learning for Accelerated MRI: Benefits of Diverse Training Data
2023	15	–	Learning High-Order Relationships of Brain Regions
2023	8	–	Contextualized Policy Recovery: Modeling and Interpreting Medical Decisions with Adaptive Imitation Learning

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of the 12 works indexed there, newest year first.

CVPR: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	68	265.4	Enhanced Contrastive Learning with Multi-view Longitudinal Data for Chest X-ray Report Generation
2024	84	47.9	Towards Generalizable Tumor Synthesis
2024	81	14.6	Rethinking Diffusion Model for Multi-Contrast MRI Super-Resolution
2024	73	13.8	MemSAM: Taming Segment Anything Model for Echocardiography Video Segmentation
2023	243	20.2	Dynamic Graph Enhanced Contrastive Learning for Chest X-Ray Report Generation
2023	222	75.5	Ambiguous Medical Image Segmentation Using Diffusion Models
2023	217	18.7	METransformer: Radiology Report Generation by Transformer with Multiple Learnable Expert Tokens
2023	164	21.9	MCF: Mutual Correction Framework for Semi-Supervised Medical Image Segmentation
2023	133	14.4	KiUT: Knowledge-injected U-Transformer for Radiology Report Generation
2023	117	67.2	Structure-Aware Sparse-View X-Ray 3D Reconstruction
2023	103	11.0	Label-Free Liver Tumor Segmentation
2023	75	14.1	Learning Federated Visual Prompt in Null Space for MRI Reconstruction
2024	22	–	M3-UDA: A New Benchmark for Unsupervised Domain Adaptive Fetal Cardiac Structure Detection ★

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 13 most-cited of the 57 works indexed there, newest year first.

MICCAI: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	208	1.1	MedVLM-R1: Incentivizing Medical Reasoning Capability of Vision-Language Models (VLMs) via Reinforcement Learning
2024	133	–	CT2Rep: Automated Radiology Report Generation for 3D Medical Imaging
2024	102	–	Anatomically-Controllable Medical Image Generation with Segmentation-Guided Diffusion Models
2024	87	–	MedCLIP-SAM: Bridging Text and Image Towards Universal Medical Image Segmentation
2023	462	–	MedNeXT: Transformer-driven Scaling of ConvNets for Medical Image Segmentation
2023	128	–	CoLa-Diff: Conditional Latent Diffusion Model for Multi-Modal MRI Synthesis
2023	105	–	DiffMIC: Dual-Guidance Diffusion Network for Medical Image Classification
2023	97	13.9	Beyond Adapting SAM: Towards End-to-End Ultrasound Image Segmentation via Auto Prompting
2023	95	–	Self-Supervised MRI Reconstruction with Unrolled Diffusion Models
2023	83	–	Make-A-Volume: Leveraging Latent Diffusion Models for Cross-Modality 3D Brain MRI Synthesis
2023	72	19.0	BerDiff: Conditional Bernoulli Diffusion Model for Medical Image Segmentation
2023	71	–	Ariadne's Thread: Using Text Prompts to Improve Segmentation of Infected Areas from Chest X-ray Images
2023	–	–	An automated pipeline for quantitative T2* fetal body MRI and segmentation at low field *
2023	–	–	ASC: Appearance and Structure Consistency for Unsupervised Domain Adaptation in Fetal Brain MRI Segmentation *
2023	–	–	Dual Conditioned Diffusion Models for Out-Of-Distribution Detection: Application to Fetal Ultrasound Videos *
2023	–	–	Improving Automatic Fetal Biometry Measurement with Swoosh Activation Function *
2023	–	–	Punctate White Matter Lesion Segmentation in Preterm Infants Powered by Counterfactually Generative Learning *
2023	–	–	Simulation-based parameter optimization for fetal brain MRI super-resolution reconstruction *
2023	–	–	SurfFlow: A Flow-Based Approach for Rapid and Accurate Cortical Surface Reconstruction from Infant Brain MRI *
2023	–	–	UM-CAM: Uncertainty-weighted Multi-resolution Class Activation Maps for Weakly-supervised Fetal Brain Segmentation *
2024	–	–	Cycle-consistent Learning for Fetal Cortical Surface Reconstruction *
2024	–	–	Efficient and Gender-adaptive Graph Vision Mamba for Pediatric Bone Age Assessment *
2024	–	–	Fetal MRI Reconstruction by Global Diffusion and Consistent Implicit Representation *
2024	–	–	Geometric Transformation Uncertainty for Improving 3D Fetal Brain Pose Prediction from Freehand 2D Ultrasound Videos *
2024	–	–	Improving cross-domain brain tissue segmentation in fetal MRI with synthetic data *
2024	–	–	Intrapartum Ultrasound Image Segmentation of Pubic Symphysis and Fetal Head Using Dual Student-Teacher Framework with CNN-ViT Collaborative Learning *
2024	–	–	Pose-GuideNet: Automatic Scanning Guidance for Fetal Head Ultrasound from Pose Estimation *
2024	–	–	Super-Field MRI Synthesis for Infant Brains Enhanced by Dual Channel Latent Diffusion *
2025	–	–	Accurate and Efficient Fetal Birth Weight Estimation from 3D Ultrasound *
2025	–	–	Adaptive Frame Selection for Gestational Age Estimation from Blind Sweep Fetal Ultrasound Videos *
2025	–	–	Flexibly Distilled 3D Rectified Flow with Anatomical Constraints for Developmental Infant Brain MRI Prediction *
2025	–	–	Fusing Radiomic Features with Deep Representations for Gestational Age Estimation in Fetal Ultrasound Images *
2025	–	–	General Methods Make Great Domain-specific Foundation Models: A Case-study on Fetal Ultrasound *
2025	–	–	Influence of Classification Task and Distribution Shift Type on OOD Detection in Fetal Ultrasound *
2025	–	–	Latent Motion Profiling for Annotation-free Cardiac Phase Detection in Adult and Fetal Echocardiography Videos *
2025	–	–	Robust Fetal Pose Estimation across Gestational Ages via Cross-Population Augmentation *
2025	–	–	Self-supervised Normality Learning and Divergence Vector-guided Model Merging for Zero-shot Congenital Heart Disease Detection in Fetal Ultrasound Videos *
2025	–	–	You Can Detect It: Fetal Biometric Estimation Using Ellipse Detection *

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 38 most-cited of the 521 works indexed there, newest year first.

MIDL: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2024	14	–	Imbalance-aware loss functions improve medical image classification
2023	78	–	Patched Diffusion Models for Unsupervised Anomaly Detection in Brain MRI
2023	39	–	MMCFormer: Missing Modality Compensation Transformer for Brain Tumor Segmentation
2023	29	–	Medical diffusion on a budget: textual inversion for medical image generation
2023	23	–	MProtoNet: A Case-Based Interpretable Model for Brain Tumor Classification with 3D Multi-parametric Magnetic Resonance Imaging
2023	22	–	Diffusion Models for Contrast Harmonization of Magnetic Resonance Images
2023	20	–	Exploring Image Augmentations for Siamese Representation Learning with Chest X-Rays
2023	19	–	3D Medical Axial Transformer: A Lightweight Transformer Model for 3D Brain Tumor Segmentation
2023	19	–	Vision-Language Modelling For Radiological Imaging and Reports In The Low Data Regime
2023	14	–	E(3) x SO(3)-Equivariant Networks for Spherical Deconvolution in Diffusion MRI
2023	14	–	Amortized Normalizing Flows for Transcranial Ultrasound with Uncertainty Quantification
2023	13	–	Selective experience replay compression using coresets for lifelong deep reinforcement learning in medical imaging
2023	–	–	FUSQA: Fetal Ultrasound Segmentation Quality Assessment *
2024	–	–	Joint Motion Estimation with Geometric Deformation Correction for Fetal Echo Planar Images Via Deep Learning *
2025	–	–	GAMBAS: Generalised-Hilbert Mamba for Super-resolution of Paediatric Ultra-Low-Field MRI *
2025	–	–	FERN: A Fetal Echocardiography Registration Network for 2D-to-3D Alignment *
2026	–	–	WristMIR: Coarse-to-Fine Region-Aware Retrieval of Pediatric Wrist Radiographs with Radiology Report-Driven Learning *
2026	–	–	Ultra-ECP: Ellipse-Constrained and Point-Robust Foundation Model Adaptation for Fetal Cardiac Ultrasound Segmentation *
2026	–	–	Calibration-Aware Semi-Supervised Fetal Head Segmentation with Boundary-Positive Contrast *
2026	–	–	SuD-CoTAN: Sulcal Depth-guided Anatomically Consistent Fetal Cortical Surface Reconstruction *
2026	–	–	Orientation-Aware Diffusion Super-Resolution for 3T-Like Fetal MRI from Routine 1.5T Scans *

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 21 most-cited of the 62 works indexed there, newest year first.

RSNA: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2024	452	113.7	Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 Update.
2024	324	26.4	Chatbots and Large Language Models in Radiology: A Practical Primer for Clinical and Research Applications.
2024	215	58.7	The Image Biomarker Standardization Initiative: Standardized Convolutional Filters for Reproducible Radiomics and Enhanced Clinical Insights.
2023	919	29.4	ChatGPT and Other Large Language Models Are Double-edged Swords.
2023	419	13.5	Performance of ChatGPT on a Radiology Board-style Examination: Insights into Current Strengths and Limitations.
2023	392	33.2	Deep Learning Image Reconstruction for CT: Technical Principles and Clinical Prospects.
2023	282	9.1	Automation Bias in Mammography: The Impact of Artificial Intelligence BI-RADS Suggestions on Reader Performance.
2023	273	44.4	Potential of ChatGPT and GPT-4 for Data Mining of Free-Text CT Reports on Lung Cancer.
2023	265	37.0	MRI-based Quantification of Intratumoral Heterogeneity for Predicting Treatment Response to Neoadjuvant Chemotherapy in Breast Cancer.
2023	259	47.2	How AI Responds to Common Lung Cancer Questions: ChatGPT vs Google Bard.
2023	258	35.5	A Guide to Cross-Validation for Artificial Intelligence in Medical Imaging.
2023	225	36.2	Predicting Microvascular Invasion in Hepatocellular Carcinoma Using CT-based Radiomics Model.

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of the 616 works indexed there, newest year first.

SPR: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	18	4.6	AI implementation in pediatric radiology for patient safety: a multi-society statement from the ACR, ESPR, SPR, SLARP, AOSPR, SPIN *
2025	17	27.9	Scanner-based real-time three-dimensional brain + body slice-to-volume reconstruction for T2-weighted 0.55-T low-field fetal magnetic resonance imaging *
2024	24	2.0	Capability of multimodal large language models to interpret pediatric radiological images *
2024	15	–	Artificial Intelligence vs. Doctors: Diagnosing Necrotizing Enterocolitis on Abdominal Radiographs
2023	49	1.6	Detecting pediatric wrist fractures using deep-learning-based object detection *
2023	44	1.2	Artificial intelligence to identify fractures on pediatric and young adult upper extremity radiographs *
2023	43	9.7	Deeplasia: deep learning for bone age assessment validated on skeletal dysplasias *
2023	42	1.2	Comparison of diagnostic performance of a deep learning algorithm, emergency physicians, junior radiologists and senior radiologists in the detection of appendicular fractures in children
2023	35	1.0	Artificial intelligence-based detection of paediatric appendicular skeletal fractures: performance and limitations for common fracture types and locations *
2023	31	1.0	The unintended consequences of artificial intelligence in paediatric radiology *
2023	23	4.0	Artificial Intelligence in Paediatric Tuberculosis *
2023	20	4.2	Innovative advances in pediatric radiology: computed tomography reconstruction techniques, photon-counting detector computed tomography, and beyond *

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of the 120 works indexed there, newest year first.

MICCAI: pediatric radiology AI works (1/2)

26 pediatric radiology-AI papers accepted 2023–2026 (21 fetal, 4 infant / neonatal, 1 pediatric). Meetings with a published paper list: 2023, 2024, 2025.

Year	Population	Paper
2025	Fetal	Accurate and Efficient Fetal Birth Weight Estimation from 3D Ultrasound
2025	Fetal	Adaptive Frame Selection for Gestational Age Estimation from Blind Sweep Fetal Ultrasound Videos
2025	Infant / neonatal	Flexibly Distilled 3D Rectified Flow with Anatomical Constraints for Developmental Infant Brain MRI Prediction
2025	Fetal	Fusing Radiomic Features with Deep Representations for Gestational Age Estimation in Fetal Ultrasound Images
2025	Fetal	General Methods Make Great Domain-specific Foundation Models: A Case-study on Fetal Ultrasound
2025	Fetal	Influence of Classification Task and Distribution Shift Type on OOD Detection in Fetal Ultrasound
2025	Fetal	Latent Motion Profiling for Annotation-free Cardiac Phase Detection in Adult and Fetal Echocardiography Videos
2025	Fetal	Robust Fetal Pose Estimation across Gestational Ages via Cross-Population Augmentation
2025	Fetal	Self-supervised Normality Learning and Divergence Vector-guided Model Merging for Zero-shot Congenital Heart Disease Detection in Fetal Ultrasound Videos
2025	Fetal	You Can Detect It: Fetal Biometric Estimation Using Ellipse Detection
2024	Fetal	Cycle-consistent Learning for Fetal Cortical Surface Reconstruction
2024	Pediatric	Efficient and Gender-adaptive Graph Vision Mamba for Pediatric Bone Age Assessment
2024	Fetal	Fetal MRI Reconstruction by Global Diffusion and Consistent Implicit Representation
2024	Fetal	Geometric Transformation Uncertainty for Improving 3D Fetal Brain Pose Prediction from Freehand 2D Ultrasound Videos

Source: the meeting's accepted-paper list; a title counts when it names an imaging modality, an AI method and a pediatric term (fetal included). Titles only, so pediatric work that does not say so in its title is missed.

MICCAI: pediatric radiology AI works (2/2)

26 pediatric radiology-AI papers accepted 2023–2026 (21 fetal, 4 infant / neonatal, 1 pediatric). Meetings with a published paper list: 2023, 2024, 2025.

Year	Population	Paper
2024	Fetal	Improving cross-domain brain tissue segmentation in fetal MRI with synthetic data
2024	Fetal	Intrapartum Ultrasound Image Segmentation of Pubic Symphysis and Fetal Head Using Dual Student-Teacher Framework with CNN-ViT Collaborative Learning
2024	Fetal	Pose-GuideNet: Automatic Scanning Guidance for Fetal Head Ultrasound from Pose Estimation
2024	Infant / neonatal	Super-Field MRI Synthesis for Infant Brains Enhanced by Dual Channel Latent Diffusion
2023	Fetal	An automated pipeline for quantitative T2* fetal body MRI and segmentation at low field
2023	Fetal	ASC: Appearance and Structure Consistency for Unsupervised Domain Adaptation in Fetal Brain MRI Segmentation
2023	Fetal	Dual Conditioned Diffusion Models for Out-Of-Distribution Detection: Application to Fetal Ultrasound Videos
2023	Fetal	Improving Automatic Fetal Biometry Measurement with Swoosh Activation Function
2023	Infant / neonatal	Punctate White Matter Lesion Segmentation in Preterm Infants Powered by Counterfactually Generative Learning
2023	Fetal	Simulation-based parameter optimization for fetal brain MRI super-resolution reconstruction
2023	Infant / neonatal	SurfFlow: A Flow-Based Approach for Rapid and Accurate Cortical Surface Reconstruction from Infant Brain MRI
2023	Fetal	UM-CAM: Uncertainty-weighted Multi-resolution Class Activation Maps for Weakly-supervised Fetal Brain Segmentation

Source: the meeting's accepted-paper list; a title counts when it names an imaging modality, an AI method and a pediatric term (fetal included). Titles only, so pediatric work that does not say so in its title is missed.

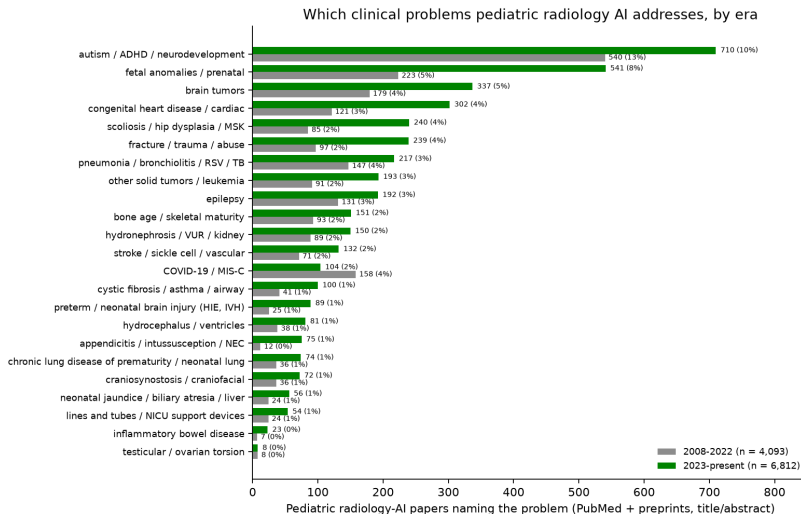
MIDL: pediatric radiology AI works

9 pediatric radiology-AI papers accepted 2023–2026 (7 fetal, 2 pediatric). Meetings with a published paper list: 2023, 2024, 2025, 2026.

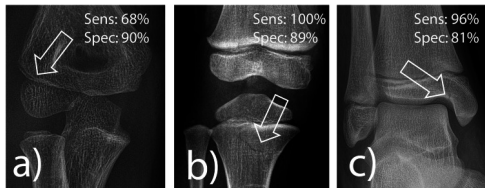
Year	Population	Paper
2026	Fetal	Calibration-Aware Semi-Supervised Fetal Head Segmentation with Boundary-Positive Contrast
2026	Fetal	Orientation-Aware Diffusion Super-Resolution for 3T-Like Fetal MRI from Routine 1.5T Scans
2026	Fetal	SuD-CoTAN: Sulcal Depth-guided Anatomically Consistent Fetal Cortical Surface Reconstruction
2026	Fetal	Ultra-ECP: Ellipse-Constrained and Point-Robust Foundation Model Adaptation for Fetal Cardiac Ultrasound Segmentation
2026	Pediatric	WristMIR: Coarse-to-Fine Region-Aware Retrieval of Pediatric Wrist Radiographs with Radiology Report-Driven Learning
2025	Fetal	FERN: A Fetal Echocardiography Registration Network for 2D-to-3D Alignment
2025	Pediatric	GAMBAS: Generalised-Hilbert Mamba for Super-resolution of Paediatric Ultra-Low-Field MRI
2024	Fetal	Joint Motion Estimation with Geometric Deformation Correction for Fetal Echo Planar Images Via Deep Learning
2023	Fetal	FUSQA: Fetal Ultrasound Segmentation Quality Assessment

Source: the meeting's accepted-paper list; a title counts when it names an imaging modality, an AI method and a pediatric term (fetal included). Titles only, so pediatric work that does not say so in its title is missed.

Which medical problems pediatric radiology AI addresses



AI fracture detection in a real pediatric emergency department

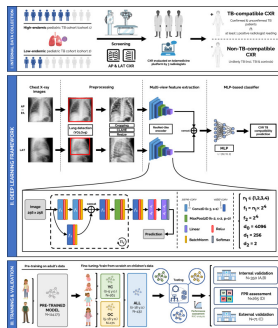


Ziegner et al., Eur Radiol 2025, figure 4: pediatric fractures the software found, with its per-region sensitivity and specificity

source: cdn.ncbi.nlm.nih.gov

- **Problem:** fractures in children are missed most often by the least experienced readers, at night, in the emergency department.
- **Data:** 1,672 consecutive radiographs of children under 18 from one tertiary pediatric emergency department, plus a selected medicolegal set (proximal tibia, radial condyle); three pediatric residents read each before and after seeing the AI.
- **Method:** a commercially available, CE-marked deep-learning fracture detector (adult-derived, pediatric extension) run stand-alone and as a second reader.
- **Result:** stand-alone sensitivity 92%, specificity 83%; 100% for proximal tibia but 68% for radial condyle fractures; residents' accuracy rose from 88% to 90%, and in 2% of cases they dropped a correct diagnosis after seeing the AI.
- **Why it matters:** the first real-life pediatric ED evaluation: the gain from a cleared adult tool is real but modest, and the elbow remains the weak spot.

pTBLightNet: pediatric tuberculosis on frontal and lateral chest radiographs



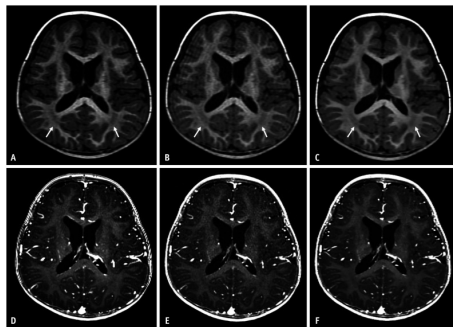
Capellan-Martin et al., Nat Commun 2025, figure 1: the multi-view

pTBLightNet pipeline

source: cdn.ncbi.nlm.nih.gov

- **Problem:** pediatric pulmonary TB has vaguer symptoms and subtler radiographs than adult TB; most affected children in high-burden settings are never diagnosed.
- **Data:** pre-trained on 114,173 adult chest radiographs, then fine-tuned and tested on 918 radiographs from three pediatric TB cohorts, with lateral views.
- **Method:** a multi-view network that scores frontal and lateral films for TB-compatible findings; separate models for under-5 and 5-18 years.
- **Result:** AUC 0.90 on internal and 0.68 on external testing; lateral views helped most in the youngest children; age-specific models matched larger undifferentiated ones.
- **Why it matters:** shows both the value of adult pre-training for a rare pediatric target and the internal-to-external drop that every pediatric deployment should expect.

Deep-learning reconstruction shortens pediatric brain MRI



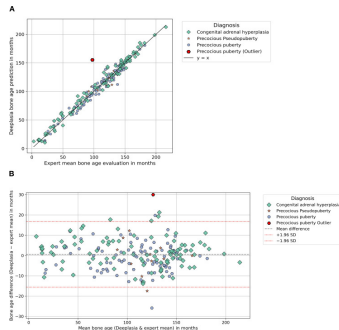
Yoo et al., Korean J Radiol 2025, figure 2: conventional (A, D) vs accelerated 3D T1 pediatric brain MRI without (B, E) and with (C, F)

deep-learning reconstruction

source: cdn.ncbi.nlm.nih.gov

- **Problem:** long 3D T1 sequences are where children move, sedation is prescribed, and scanner time is lost.
- **Data:** 46 children scanned with both the conventional and an accelerated 3D T1 protocol (pre- and post-contrast) on one 3T scanner.
- **Method:** vendor deep-learning reconstruction applied to the accelerated acquisition; noise, contrast, signal-to-noise and radiologist ratings compared with the conventional scan.
- **Result:** acquisition time cut by 29% (pre-contrast) and 41% (post-contrast); the deep-learning images had lower noise, higher SNR and CNR, better overall quality and fewer artifacts than the conventional scan.
- **Why it matters:** reconstruction AI is on the scanner today and gives children a shorter, better exam without any diagnostic model in the loop.

Deeplasia bone age in rare growth disorders, against multiple experts

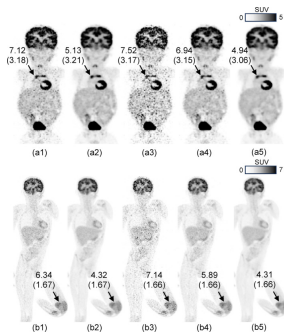


Skaf et al., Front Endocrinol 2026, figure 1: Deeplasia bone age vs the expert mean in the endocrine cohort (scatter and Bland-Altman)

source: cdn.ncbi.nlm.nih.gov

- **Problem:** bone-age models are trained on ordinary hands; the children who need bone age most have syndromes, endocrine disease or lysosomal storage disorders that distort the skeleton.
- **Data:** 1,138 hand radiographs from several centers: SHOX deficiency, Noonan, Silver-Russell, Turner, CAH, precocious puberty (cohort 1) and mucopolysaccharidoses and other storage disorders (cohort 2); two to five expert Greulich-Pyle readings each.
- **Method:** the open-source Deeplasia system run unchanged; each human rater and the model scored against the remaining experts.
- **Result:** mean absolute error 6.0 months (cohort 1) and 7.1 months (storage disorders), 1-year accuracy 90% and 81%; the model beat every individual human rater.
- **Why it matters:** an open pediatric model validated on the hardest cases, by an outside group, on external data: the template for how a children's hospital should test any tool.

Pediatric PET at one fifth of the counts with patch-based deep learning



Han et al., EJNMMI Phys 2026, figure 5: pediatric whole-body FDG

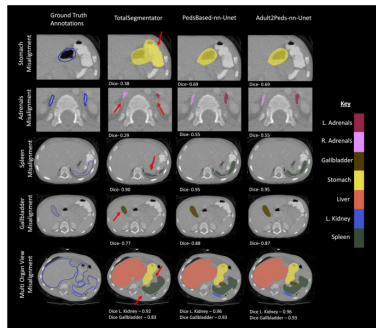
PET, standard 90 s vs 20 s per bed, with and without patch-based

deep-learning denoising

source: cdn.ncbi.nlm.nih.gov

- **Problem:** PET dose and scan time in children are set by noise; conventional denoising networks need large training sets that pediatric PET does not have.
- **Data:** one high-quality pediatric exam for training; a NEMA phantom and 88 clinical whole-body FDG exams (under 12 years) with list-mode data truncated from 90 s to 20 s per bed.
- **Method:** a patch-based network that learns local structure from a single exam and denoises the reduced-count images; blinded observer study and SUV analysis.
- **Result:** 20-second images rated equal or better than the standard 90-second images in more than 85% of cases, with resolution, contrast and SUV preserved.
- **Why it matters:** either a 4.5x shorter scan or about 22% of today's injected activity, from a method trainable on a single local exam.

Children are not small adults: an adult CT segmentation model in children



Chatterjee et al., J Imaging Inform Med 2025, figure 7: pediatric CT

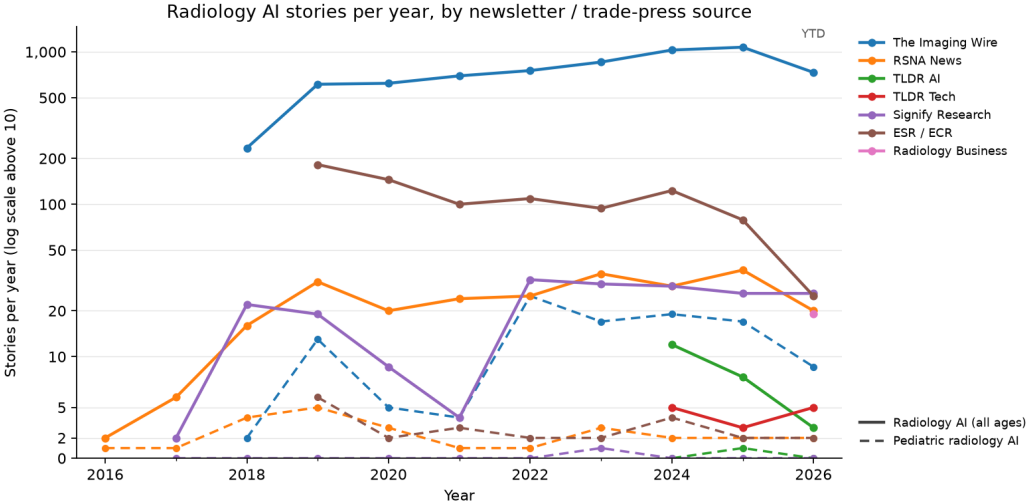
organ labels, ground truth vs adult-trained TotalSegmentator vs

pediatric-trained models

source: cdn.ncbi.nlm.nih.gov

- **Problem:** the most-used open CT organ segmentation model (TotalSegmentator) was trained on adults; nobody had measured how it behaves in children.
- **Data:** external adult ($n = 300$) and pediatric ($n = 359$) abdominal CT sets with expert organ labels.
- **Method:** TotalSegmentator scored on both sets; then a pediatric-only nnU-Net and an adult model fine-tuned on pediatric cases.
- **Result:** mean Dice fell from 0.81 in adults to 0.73 in children, worst for the adrenals (0.69 to 0.41 and 0.35), duodenum and pancreas; pediatric training or fine-tuning recovered the loss.
- **Why it matters:** the clearest quantitative case for local pediatric validation and fine-tuning of adult-derived tools before clinical use.

Newsletters — radiology AI stories per year, by source



Pediatric radiology AI in the trade press, 2023 (23 stories)

Date	Newsletter	Story	Paper
2023-01-08	The Imaging Wire	Gestational Age AI	
2023-03-01	The Imaging Wire	Fetal Ultrasound AI Generalizability	Sendra-Balcells et al., 2023 (Scientific Reports)
2023-03-08	ESR / ECR	ECR 2023 – A true celebration of the cycle of life.	
2023-03-17	The Imaging Wire	When Sao Paolo's Diagnosticos da America (DASA, the world's fourth largest diagn	
2023-04-07	The Imaging Wire	Ultrasound Spots Breech Pregnancies	Knights et al., 2023 (PLOS Medicine)
2023-04-07	The Imaging Wire	When Sao Paolo's Diagnosticos da America SA (DASA, the world's fourth largest di	
2023-04-12	Signify Research	Where do opportunities exist?	
2023-04-12	RSNA News	RSNA Webinar to Examine Impact of Large Language Models	
2023-05-07	The Imaging Wire	New C-Arm Launched	
2023-06-07	The Imaging Wire	BrightHeart's Fetal Heart AI	
2023-06-12	RSNA News	Understanding the Imaging Spectrum of non-Hodgkin Lymphoma in Children, Adolescents and Young Adults	
2023-06-19	The Imaging Wire	Better Together at SIIM	
2023-06-25	The Imaging Wire	AI Detects Child Abuse on CT	
2023-07-07	RSNA News	Survivors of Childhood Cancer: Risk of Future Disease, Screening Needs and the Role of Radiology	
2023-08-02	The Imaging Wire	MRI Radiomics Hones in on Crohn's	Liu et al., 2024 (American Journal of Roentgenology)
2023-08-02	The Imaging Wire	Sonio Gets FDA Nod for Fetal AI	
2023-09-20	The Imaging Wire	GE Lands \$44M AI Ultrasound Grant	
2023-10-08	The Imaging Wire	ChatGPT Reports for Patients	Jeblick et al., 2023 (European Radiology)
2023-11-08	The Imaging Wire	Philips to Expand AI-Powered POCUS	
2023-11-08	The Imaging Wire	Stories of Success in Cloud and AI	
2023-11-12	The Imaging Wire	Uneven Success Against Breast Cancer Pediatric Radiation	
2023-11-26	The Imaging Wire	AI of MRI Diagnoses Autism	
2023-12-11	ESR / ECR	Revolutionising paediatric radiology: the future impact of AI	

Pediatric radiology AI in the trade press, 2024 (25 stories)

Date	Newsletter	Story	Paper
2024-01-10	The Imaging Wire	AI Models Go Head-to-Head in Project AIR Study	van Leeuwen et al., 2024 (Radiology)
2024-01-15	ESR / ECR	Hidden health risks of endocrine disruptors?	
2024-01-17	The Imaging Wire	Echo AI's RHD Impact	Brown et al., 2024 (Journal of the American Heart Association)
2024-01-20	The Imaging Wire	Us2.ai	
2024-02-07	The Imaging Wire	Meet Gleamer at ECR 2024	
2024-03-20	The Imaging Wire	AI Helps Residents Detect Fractures	
2024-04-22	The Imaging Wire	GE Launches US Scanners	
2024-05-19	The Imaging Wire	Early Preeclampsia Risk Screening	Guerby et al., 2024 (Hypertension)
2024-06-29	The Imaging Wire	FDA Clears New Clarius Prenatal App	
2024-07-01	ESR / ECR	AI applications in musculoskeletal imaging	
2024-07-10	RSNA News	Experts Outline Considerations to Deploy AI in Radiology	
2024-08-14	The Imaging Wire	Cardiac Ultrasound Scanner Cleared	
2024-08-25	The Imaging Wire	When Follow-Up Falls Short Pediatric CT Use	
2024-08-25	The Imaging Wire	MAUI Emerges from Stealth	
2024-09-04	The Imaging Wire	Samsung Finalizes Sonio Acquisition	
2024-09-08	The Imaging Wire	AZmed AI Extended to Kids	
2024-10-03	ESR / ECR	On Artificial Intelligence: An interview with Susan Shelmerdine	
2024-10-06	The Imaging Wire	Fracture AI Improves Over Time	
2024-10-10	The Imaging Wire	Low-Dose CT in Kids Hinton Nabs Nobel, Cancels MRI	
2024-10-10	The Imaging Wire	Low-Dose CT Confounds CAD in Kids	Hardie et al., 2025 (American Journal of Roentgenology)
2024-10-17	ESR / ECR	On Artificial Intelligence: An interview with Brendan Kelly	
2024-11-19	RSNA News	Incorrect AI Advice Influences Diagnostic Decisions	
2024-11-25	The Imaging Wire	Fetal Ultrasound AI Cleared	
2024-12-02	The Imaging Wire	Milvue Joins deepc AI Platform	
2024-12-15	The Imaging Wire	Researchers Study AI for Pediatric Fractures	

Pediatric radiology AI in the trade press, 2025 (22 stories)

Date	Newsletter	Story	Paper
2025-01-08	The Imaging Wire	AI for Pediatric Fracture Detection	
2025-01-08	The Imaging Wire	Children's National taps Microsoft to prototype AI tools .	
2025-01-29	The Imaging Wire	Samsung Launches OB/GYN Ultrasound	
2025-02-02	The Imaging Wire	Ultrasound AI Detects Fetal Defects	
2025-02-06	ESR / ECR	Philips at #ECR2025: New AI-enabled systems, intelligent software, and imaging cloud services streamline radiology workflows, advance clinical insights	
2025-02-14	ESR / ECR	Learn more about FUJIFILM's latest innovations at ECR!	Iwan et al., 2020 (European Radiology)
2025-02-27	The Imaging Wire	AI Detects Pediatric Epilepsy	
2025-03-04	TLDR AI	AI to diagnose invisible brain abnormalities in children with epilepsy (4 minute read)	
2025-04-02	The Imaging Wire	AZmed Gets New Chest AI Clearances	
2025-04-13	The Imaging Wire	GE's Cincinnati Pediatric Partnership	
2025-05-14	The Imaging Wire	Philips Partners with NVIDIA on AI for MRI	
2025-05-18	The Imaging Wire	New GE Clearances	
2025-05-20	RSNA News	Radiologists Share Tips to Prevent AI Bias	
2025-06-01	The Imaging Wire	BrightHeart's 3rd FDA Clearance	
2025-07-03	The Imaging Wire	DeepEcho Gets FDA Nod for Fetal AI	
2025-07-10	RSNA News	How Is AI Being Used in Daily Neuroradiology Practice?	
2025-08-03	The Imaging Wire	Radiologist Pay Jumps Nearly 8% in New Survey	
2025-08-13	The Imaging Wire	AI-Assisted Fracture Detection	
2025-08-27	The Imaging Wire	Sonio Launches New Maternal-Fetal AI Options	
2025-10-01	The Imaging Wire	Carebot Expands AI Portfolio	
2025-10-08	The Imaging Wire	Reducing CT Radiation Dose System-Wide	
2025-11-09	The Imaging Wire	AI Detects Pediatric Elbow Fractures	

Pediatric radiology AI in the trade press, 2026 (year to date) (13 stories)

Date	Newsletter	Story	Paper
2026-01-07	The Imaging Wire	BrightHeart Raises EUR 11M	
2026-01-28	The Imaging Wire	AHA Publishes New Stroke Guidelines	Prabhakaran et al., 2026 (Stroke)
2026-01-28	ESR / ECR	Guerbet: Innovation and French Cooperation in the Spotlight at ECR 2026	
2026-02-01	The Imaging Wire	DL-Based CT Reduces Radiation Dose	
2026-02-08	The Imaging Wire	GE Bolsters Ultrasound with Fetal AI	
2026-02-26	ESR / ECR	MRI Unveiling at the	
2026-03-29	The Imaging Wire	Ultrasound AI Gets Breakthrough Nod	
2026-05-10	The Imaging Wire	Pediatric MRI Safety, RTs and Remote Radiology, and PSMA-PET	
2026-06-22	RSNA News	Researchers Use AI to Establish Body Charts for CT Imaging Across the Adult Lifespan	
2026-06-28	The Imaging Wire	BrightHeart's Bright Future	Zelop et al., 2025 (Obstetrics & Gynecology)
2026-07-12	The Imaging Wire	AI's Rise in Pediatric Imaging	Shah et al., 2026 (European Radiology)
2026-08-16	The Imaging Wire	CT vs. MRI for Pediatric TBI, Helium Vulnerability, and Ditching the Disk	
2026-08-25	RSNA News	AI Challenges Fuel Innovation Beyond the Leaderboard	

Commercial software with a pediatric angle

Vendor	Selected products	Functions	Pediatric / regulatory scope	Company-wide FDA AI entries (n; years)
Gleamer	BoneView	X-ray fracture detection; international versions also flag dislocations, effusions and focal bone lesions	US fracture aid: ages 2–21 for specified extremity views; other anatomy adult-only (2023)	4 (2022–2025)
AZmed	Rayvolve suite	MSK / chest finding detection; orthopedic measurements; GP bone age (separate modules)	US fracture indication age 2+ (2024); AZboneage is CE-marked; other eligibility varies by module	4 (2022–2025)
Visiana	BoneXpert	Hand X-ray bone age (GP / TW3), Bone Health Index; associated adult-height prediction tool	Designed for children; CE-marked; US research use only	not listed
16 Bit	Physis	Hand X-ray bone-age estimation, GP atlas matching and report assistance	Pediatric tool; no FDA authorization identified. Company's FDA entry is Rho (low bone density, age 50+)	1 (2024)
Ever Fortune.AI	EFAI Bonesuite XR Bone Age Pro	GP bone-age quantification from PA left-hand / wrist radiographs	FDA-cleared 2024 for ages 2–16 (K234042)	11 (2022–2026)
BrightHeart	Fetal EchoScan; View Classifier	EchoScan: flags 8 findings suspicious for fetal CHD; View Classifier: standard views / exam completeness	Prenatal: EchoScan, 2nd trimester; View Classifier, 2nd / 3rd; maternal age 18+	5 (2024–2025)
GE HealthCare / Canon / Siemens / Philips	TrueFidelity, AiCE, Deep Resolve, Precise Image	CT: TrueFidelity / AiCE / Precise Image (noise, dose); MRI: AiCE / Deep Resolve (noise, resolution, scan time)	Pediatric applications documented; age / anatomy / protocol depend on scanner and software version	284 (2014–2026)
Subtle Medical / AIRS Medical	SubtleMR, SwiftMR	MRI image post-processing: denoising / sharpening to support shorter acquisitions or better image quality	No adult-only restriction in reviewed US indications; SwiftMR 2026 validation includes ages 0–21	13 (2018–2026)
Qure.ai	qXR / qER families	Chest X-ray detection / triage, TB screening, tube / heart measurements; head CT triage / quantification and CTA LVO	US eligibility varies by module; vendor reports CE-marked qXR TB use across ages 0–15	9 (2020–2026)
Aidoc / Viz.ai	BriefCase, Viz LVO / ICH	Aidoc: CT / CTA triage (e.g., ICH, LVO, PE, pneumothorax, fractures, free gas); Viz: LVO / ICH alerts, care coordination	Age / anatomy eligibility is module-specific; company totals do not establish pediatric indications	43 (2018–2026)

Counts = company-wide radiology entries in the FDA AI-enabled device list (saved 2026-09-03); grouped vendors are summed. Includes versions and unrelated products, not pediatric-clearance counts; parentheses = decision-year range. The list is not exhaustive and may lag new authorizations.

Functions / labeling checked 2026-09-16; features vary by market and version. GP = Greulich–Pyle; TW3 = Tanner–Whitehouse 3; CHD = congenital heart disease.

FDA pediatric-use inventory: scope

- Screened all 1,230 Radiology-panel records in the FDA AI-device CSV downloaded 2026-09-16.
- 283 records had pediatric/fetal keywords; 230 are label-positive candidates.
- This is exhaustive for the dated FDA-list snapshot, not for the commercial market: FDA says its AI list is not comprehensive. Repeated submissions are retained so the result is auditable.
- Candidate status is a keyword screen of linked decision-summary PDFs. Confirm the current authorization and full labeling before purchase.

Source: `data/processed/fda_pediatric_inventory.json`; evidence pages are retained per submission.

FDA pediatric-use inventory: label-positive candidates (1–20 of 230)

Submission	Date	Company	Device	Code
DEN250007	02/11/2020	Ultrasound AI	Delivery Date AI	SHE
K163253	01/05/2017	ARTERYS INC.	Arterys Cardio DL	LLZ
K172356	09/15/2017	Verathon Incorporated	BladderScan Prime PLUS System	IYO
K172385	09/14/2017	Clarius Mobile Health Corp.	Clarius Ultrasound System	IYN
K180799	05/14/2018	Clarius Mobile Health Corp.	Clarius Ultrasound Scanner	IYN
K181574	07/10/2018	EchoNous, Inc.	Uscan	IYO
K181685	10/25/2018	GE Medical Systems Ultrasound and Primary Care Diagnostics	Vivid E80, Vivid E90, Vivid E95	IYN
K191171	11/13/2019	Ultrasonics Ltd	EchoGo Core	QIH
K191278	11/19/2019	MultiModal Imaging Services Corporation (dba HealthLytix)	RSI-MRI+	LLZ
K192107	08/29/2019	Clarius Mobile Health Corp.	Clarius Ultrasound Scanner	IYN
K192377	02/10/2020	Varian Medical Systems, Inc.	Ethos Treatment Management, Ethos Treatment Planning, Ethos Radiotherapy System, Halcyon	IYE
K193518	03/26/2020	EchoNous, Inc.	KOSMOS	IYN
K200232	06/23/2020	DiA Imaging Analysis Ltd	LVivo Software Application	QIH
K200356	06/11/2020	Medo.ai	MEDO ARIA	QIH
K200980	06/11/2020	Butterfly Network, Inc.	Auto 3D Bladder Volume Tool	IYO
K201012	05/01/2020	Philips Ultrasound, Inc.	EPIQ Diagnostic Ultrasound System Series, Affiniti Diagnostic Ultrasound System Series, CX50 Diagnostic Ultrasound System, Sparq Diagnostic Ultrasound System, Lumify Diagnostic Ultrasound System	IYN
K201039	12/07/2020	Resonance Health Analysis Services Pty Ltd	HepaFat-AI	LNH
K201632	08/14/2020	TOMTEC Imaging Systems GmbH	TOMTEC-ARENA	LLZ
K210426	05/13/2021	Samsung Medison CO., LTD.	HS40 Diagnostic Ultrasound System	IYN
K210556	04/21/2021	Genesis Software Innovations	Preview Shoulder	QIH

FDA pediatric-use inventory: label-positive candidates (21–40 of 230)

Submission	Date	Company	Device	Code
K210760	01/14/2020	Philips Medical Systems Nederland, B.V.	Precise Image	JAK
K211597	09/08/2020	Philips Medical Systems	EPIQ Series Diagnostic Ultrasound System, Affiniti Series Diagnostic Ultrasound System	IYN
K211611	09/30/2020	CASIS Cardiac Simulation & Imaging Software	QIR Suite	QIH
K211824	09/09/2020	Samsung Medical Co., Ltd.	HERA W9, HERA W10 Diagnostic Ultrasound System	IYN
K211945	09/08/2020	Samsung Medical Co., LTD.	V8 Diagnostic Ultrasound System	IYN
K211966	05/06/2020	Medviso AB	Segment 3DPrint	LLZ
K212012	01/19/2020	Cycle Clarity	Follicle Clarity	QIH
K212100	08/06/2020	EchoNous, Inc.	Kosmos	IYN
K212265	11/16/2020	Shenzhen Mindray Bio-Medical Electronics Co.,LTD	TEX20/TEX20 Pro/TEX20S/TEX20T/TEX20 Exp/TEX20 Elite Diagnostic Ultrasound System, TEX10/ TEX10 Pro/TEX10S/TEX10T/TEX10 Exp/TEX10 Elite/TE X/TE X Lite Diagnostic Ultrasound System	IYN
K212269	09/17/2020	Canon Inc.	Intelligent NR	MOB
K212336	11/17/2020	Omega Medical Imaging, LLC	Soteria.AI	OWB
K212616	12/16/2020	Koios Medical, Inc.	Koios DS	POK
K212654	02/04/2020	OXOS Medical, Inc.	Micro C Medical Imaging System, M01	IZL
K212704	09/24/2020	Philips Medical Systems	Philips EPIQ Diagnostic Ultrasound System, Philips Affiniti Diagnostic Ultrasound System	IYN
K213272	03/31/2020	Formus Labs, Ltd	Formus Hip	QIH
K213436	11/15/2020	Clarius Mobile Health Corp.	Clarius Ultrasound Scanner	IYN
K213544	01/06/2020	TOMTEC Imaging Systems GmbH	TOMTEC-ARENA	QIH
K213689	02/17/2020	GE Healthcare	Voluson P6, Voluson P8	IYN
K213886	04/26/2020	Aidoc Medical, Ltd.	BriefCase	QAS
K214043	03/14/2020	Aidoc Medical, Ltd.	BriefCase	QFM

FDA pediatric-use inventory: label-positive candidates (41–60 of 230)

Submission	Date	Company	Device	Code
K220043	04/05/2022	Samsung Medison Co., LTD	HERA W10 Diagnostic Ultrasound System, HERA W9 Diagnostic Ultrasound System	IYN
K220068	03/31/2023	Butterfly Network, Inc.	Butterfly iQ/iQ+ Ultrasound System	IYO
K220358	06/06/2022	GE Medical Systems Ultrasound and Primary Care Diagnostics,	Voluson Expert 22, Voluson Expert 20, Voluson Expert 18	IYN
K220497	06/23/2022	Smart Soft Healthcare AD	CoLumbo	QIH
K220709	10/07/2022	Aidoc Medical, Ltd.	BriefCase	QAS
K220813	06/17/2022	TheraPanacea	ART-PLAN	QKB
K220848	06/27/2022	GE Medical Systems	Venue Fit	IYN
K220851	06/27/2022	GE Medical Systems	Venue	IYN
K220975	06/29/2022	Samsung Medison Co., Ltd.	V8 Diagnostic Ultrasound System, V7 Diagnostic Ultrasound System	IYN
K221240	05/17/2022	Aidoc Medical, Ltd.	BriefCase	QAS
K221314	06/03/2022	Aidoc Medical, Ltd.	BriefCase	QAS
K221330	11/18/2022	Aidoc Medical, Ltd.	BriefCase	QAS
K221393	06/10/2022	Hyperfine, Inc.	Swoop Portable MR Imaging System	LNH
K221599	08/22/2022	Samsung Medison Co., Ltd.	HS40 Diagnostic Ultrasound System	IYN
K221923	07/28/2022	Hyperfine, Inc.	Swoop Portable MR Imaging System	LNH
K222176	03/02/2023	Gleamer	BoneView	QBS
K222277	08/26/2022	Aidoc Medical, Ltd.	BriefCase	QAS
K222329	09/28/2022	Aidoc Medical, Ltd.	BriefCase	QAS
K222406	01/23/2023	Clarius Mobile Health Corp.	Clarius AI	QIH
K222428	11/14/2022	Siemens Medical Solutions USA, Inc.	Syngo Dynamics (Version VA40F)	QIH

FDA pediatric-use inventory: label-positive candidates (61–80 of 230)

Submission	Date	Company	Device	Code
K222755	02/16/2023	Shanghai United Imaging Intelligence Co., Ltd.	uMR 680	LNH
K223268	12/16/2022	Hyperfine, Inc.	BrainInsight	QIH
K223387	02/13/2023	Samsung Medison Co., Ltd.	V8 Diagnostic Ultrasound System, V7 Diagnostic Ultrasound System	IYN
K223491	05/25/2023	GE Medical Systems, LLC	Critical Care Suite with Pneumothorax Detection AI Algorithm, Critical Care Suite 2.1, Critical Care Suite	QBS
K223771	05/04/2023	Philips Ultrasound	Lumify Diagnostic Ultrasound System	IYN
K223830	04/11/2023	BK Medical Aps	Ultrasound System 2300	IYN
K230020	02/01/2023	Aidoc Medical, Ltd.	BriefCase	QFM
K230084	04/21/2023	Samsung Medison CO., LTD.	HERA W10 Diagnostic Ultrasound System; HERA W9 Diagnostic Ultrasound System	IYN
K230112	06/13/2023	Imbio, Inc.	CAC Software	JAK
K230179	11/29/2023	Esaoite S.p.A.	6440 MyLabX90	IYN
K230208	02/22/2023	Hyperfine, Inc.	Swoop® Portable MR Imaging System	LNH
K230209	10/20/2023	Ontact Health Co., Ltd.	Sonix Health	QIH
K230346	06/20/2023	GE Medical Systems Ultrasound & Primary Care Diagnostic, LLC	Voluson SWIFT; Voluson SWIFT+	IYN
K230365	07/25/2023	Sonio	Sonio Detect	IYN
K231001	10/05/2023	DeepTek Medical Imaging Pvt Ltd	DeepTek CXR Analyzer v1.0	MYN
K231190	05/12/2023	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound System	IYN
K231677	03/06/2024	Edgecare Inc.	EdgeFlow UH10	IYO
K231764	10/23/2023	BK Medical Aps	Ultrasound System 1300	IYN
K231772	10/03/2023	Samsung Medison Co., Ltd.	V8/H8 Diagnostic Ultrasound System, V7/H7 Diagnostic Ultrasound System, V6/H6 Diagnostic Ultrasound System	IYN
K231917	01/05/2024	EOS imaging	VEA Align	QIH

FDA pediatric-use inventory: label-positive candidates (81–100 of 230)

Submission	Date	Company	Device	Code
K231965	10/30/2023	GE Medical Systems Ultrasound and Primary Care Diagnostics,	Voluson Expert 22, Voluson Expert 20, Voluson Expert 18	IYN
K231966	11/07/2023	GE Medical Systems Ultrasound and Primary care Diagnostics,	LOGIQ E10	IYN
K232083	11/13/2023	Aidoc Medical, Ltd.	BriefCase-Quantification	QIH
K232145	10/30/2023	Siemens Medical Solutions USA, Inc.	ACUSON Sequoia Diagnostic Ultrasound System, ACUSON Sequoia Select Diagnostic Ultrasound System, ACUSON Origin Diagnostic Ultrasound System	IYN
K232231	12/13/2023	Quibim S.L.	QP-Brain®	QIH
K232257	11/13/2023	Clarius Mobile Health Corp.	Clarius Bladder AI	QIH
K232384	12/15/2023	VideaHealth, Inc.	Videa Dental Assist	MYN
K232410	05/10/2024	Milvue	SmartChest	QFM
K232500	10/26/2023	Philips Ultrasound	Lumify Diagnostic Ultrasound System	IYN
K232613	02/28/2024	Innolitics, LLC	CT Cardiomegaly	QIH
K232698	01/18/2024	Software Nemotec S.L.	NemoScan	QIH
K232704	10/05/2023	Clarius Mobile Health Corp.	Clarius Ultrasound Scanner	IYN
K232751	10/30/2023	Aidoc Medical, Ltd.	BriefCase-Triage	QAS
K232760	10/06/2023	Hyperfine, Inc.	Swoop® Portable MR Imaging System®	LNH
K232808	01/04/2024	Butterfly Network, Inc.	Butterfly iQ3 Ultrasound System	IYN
K233030	03/01/2024	MRGuidance B.V	BoneMRI	QIH
K233247	05/15/2024	Heuron Co., Ltd.	Heuron ICH	QAS
K233457	07/12/2024	Hutom Inc.	RUS	QIH
K233543	05/21/2024	Siemens Medical Solutions	YSIO X.pree	KPR
K233545	11/30/2023	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound System	IYN

FDA pediatric-use inventory: label-positive candidates (101–120 of 230)

Submission	Date	Company	Device	Code
K233738	03/04/2024	Overjet, Inc	Overjet Caries Assist-Pediatric	MYN
K233788	02/13/2024	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound System, Affiniti Series Diagnostic Ultrasound System	IYN
K233955	06/14/2024	Clarius Mobile Health Corp.	Clarius OB AI	IYN
K233968	03/13/2024	Avicenna.AI	CINA-iPE	QAS
K233977	09/04/2024	Sonic Incytes	Velacur	IYO
K234042	06/07/2024	Ever Fortune.AI Co., Ltd.	EFAI Bonesuite XR Bone Age Pro Assessment System (BAP-XR-100)	QIH
K240291	04/08/2024	Ever Fortune.AI Co., Ltd.	EFAI CARDIOSUITE CTA ACUTE AORTIC SYNDROME ASSESSMENT SYSTEM	QAS
K240406	04/26/2024	Sonio	Sonio Detect	IYN
K240516	06/12/2024	Samsung Medison Co., Ltd.	RS85 Diagnostic Ultrasound System	IYN
K240582	06/25/2024	EOS imaging	VEA Align; spineEOS	QIH
K240631	06/21/2024	Samsung Medison Co., Ltd.	V8/XV8/XH8, V7/XV7/XH7, V6/XV6/XH6 Diagnostic Ultrasound System	IYN
K240704	07/22/2024	Siemens Medical Solutions, USA, Inc.	ACUSON Sequoia Diagnostic Ultrasound System, ACUSON Sequoia Select Diagnostic Ultrasound System, ACUSON Origin Diagnostic Ultrasound System, ACUSON Origin ICE Diagnostic Ultrasound System	IYN
K240793	12/16/2024	MSKai	MSKai	QIH
K240845	07/17/2024	AZmed SAS	Rayvolve	QBS
K240850	04/24/2024	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound Systems; Affiniti Series Diagnostic Ultrasound Systems	IYN
K240944	07/16/2024	Hyperfine, Inc.	Swoop® Portable MR Imaging® System	LNH
K240980	10/07/2024	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound System	IYN
K241029	10/07/2024	Verdure Imaging	SpineUs System	IYO
K241108	10/30/2024	Remedy Logic Inc.	RemedyLogic AI MRI Lumbar Spine Reader	QIH
K241112	05/15/2024	Aidoc Medical, Ltd.	BriefCase-Quantification	QIH

FDA pediatric-use inventory: label-positive candidates (121–140 of 230)

Submission	Date	Company	Device	Code
K241211	08/15/2024	Smart Soft Health-care	CoLumbo	QIH
K241331	10/01/2024	Springbok, Inc.	MuscleView	LNH
K241380	09/11/2024	Diagnoly	FETOLY-HEART	IYN
K241659	02/10/2025	Philips Ultrasound LLC	Ultrasound Workspace (UWS 6.0)	QIH
K241727	07/12/2024	Aidoc Medical, Ltd.	BriefCase-Triage	QAS
K241949	01/17/2025	Sonoscape Medical Corp.	Digital Color Doppler Ultrasound System (P60 Series)	IYN
K241971	10/11/2024	Samsung Medison Co., Ltd.	HERA Z20, R20, HERA Z30, R30 Diagnostic Ultrasound System	IYN
K242020	12/12/2024	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound System; Affiniti Series Diagnostic Ultrasound System	IYN
K242062	11/15/2024	Mycardium AI Limited	1CMR Pro	LLZ
K242168	12/20/2024	GE Medical Systems Ultrasound and Primary care Diagnostics	Voluson Expert 18; Voluson Expert 20; Voluson Expert 22	IYN
K242171	01/17/2025	Milvue	TechCare Trauma	QBS
K242203	11/22/2024	Aidoc Medical, Ltd.	BriefCase-Quantification	QIH
K242342	11/14/2024	BrightHeart	Fetal EchoScan	POK
K242444	11/27/2024	Samsung Medison Co., Ltd.	HERA W10 Diagnostic Ultrasound System; HERA W9 Diagnostic Ultrasound System	IYN
K242488	01/06/2025	Omega Medical Imaging, LLC	Soteria E-View	JAA
K242511	12/10/2024	Samsung Medison Co., Ltd.	V5 Diagnostic Ultrasound System, H5 Diagnostic Ultrasound System, XV5 Diagnostic Ultrasound System, XH5 Diagnostic Ultrasound System, V4 Diagnostic Ultrasound System, H4 Diagnostic Ultrasound System, XV4 Diagnostic Ultrasound System, XH4 Diagnostic Ultrasound System	IYN
K242523	11/19/2024	Siemens Medical Solutions USA, Inc.	ACUSON Sequoia Diagnostic Ultrasound System; ACUSON Sequoia Select Diagnostic Ultrasound System; ACUSON Origin Diagnostic Ultrasound System; ACUSON Origin ICE Diagnostic Ultrasound System	IYN
K242551	04/03/2025	Siemens Healthcare GmbH	syngo Dynamics (Version VA41D)	QIH
K242594	05/23/2025	DeepEcho	DEEPECHO	IYN
K242607	02/21/2025	ScanDiags AG	ScanDiags Ortho L-Spine MR-Q	QIH

FDA pediatric-use inventory: label-positive candidates (141–160 of 230)

Submission	Date	Company	Device	Code
K242800	11/15/2024	Philips Ultrasound LLC	The 5000 Compact Series Ultrasound Systems	IYN
K242821	02/20/2025	Ever Fortune.AI, Co., Ltd.	EFAl Chestsuite XR Malpositioned ETT Assessment System (ETT-XR-100)	QAS
K242837	10/18/2024	Aidoc Medical, Ltd.	BriefCase-Triage	QAS
K243548	12/11/2024	Aidoc Medical, Ltd.	BriefCase-Triage	QFM
K243614	02/21/2025	Sonio	Sonio Suspect	POK
K243684	05/07/2025	BrightHeart	BrightHeart View Classifier	QIH
K243702	02/12/2025	Samsung Medison Co., Ltd.	V8 Diagnostic Ultrasound System; cV8 Diagnostic Ultrasound System; V7 Diagnostic Ultrasound System; cV7 Diagnostic Ultrasound System; V6 Diagnostic Ultrasound System; cV6 Diagnostic Ultrasound System	IYN
K243779	07/01/2025	BunkerHill Health	Bunkerhill Abdominal Aortic Quantification (AAQ)	QIH
K243793	05/21/2025	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound System; Affiniti Series Diagnostic Ultrasound System	IYN
K243794	02/06/2025	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound System; Affiniti Series Diagnostic Ultrasound System	IYN
K243859	08/29/2025	Nurea	PRAEVAorta®2	QIH
K243893	05/05/2025	Pearl, Inc.	Second Opinion® Pediatric	MYN
K250177	04/10/2025	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound System; Affiniti Series Diagnostic Ultrasound System	IYN
K250236	05/30/2025	Hyperfine, Inc.	Swoop® Portable MR Imaging® System (V2)	LNH
K250237	09/15/2025	Beijing Intervision Healthcare Medical Technology Co., Ltd.	InferOperate Suite	QIH
K250248	02/14/2025	Aidoc Medical, Ltd.	BriefCase-Triage	QAS
K250367	05/28/2025	Smart Soft Healthcare AD	CoLumboX	QIH
K250370	05/20/2025	FUJIFILM Corporation	SCENARIA View Phase 5.0	JAK
K250416	04/11/2025	Galileo CDS, Inc	GBrain MRI	QIH
K250660	07/14/2025	Siemens Medical Solutions	LUMINOS Q.namix T; LUMINOS Q.namix R	OWB

FDA pediatric-use inventory: label-positive candidates (161–180 of 230)

Submission	Date	Company	Device	Code
K250670	06/30/2025	Mycardium AI Limited	EchoConfidence (USA)	QIH
K250686	07/22/2025	NeuroSpectrum Insights Corp.	GyrCalc (Version 1.0.0)	LLZ
K250886	06/18/2025	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound Systems; Affiniti Series Diagnostic Ultrasound Systems	IYN
K250914	12/18/2025	Crescom Co., Ltd.	MediAI-BA	QIH
K250959	12/22/2025	BioticsAI, Inc.	BioticsAI	IYN
K250999	07/18/2025	Samsung Medison Co., Ltd.	V8 Diagnostic Ultrasound System; cV8 Diagnostic Ultrasound System; V7 Diagnostic Ultrasound System; cV7 Diagnostic Ultrasound System; V6 Diagnostic Ultrasound System; cV6 Diagnostic Ultrasound System; V5 Diagnostic Ultrasound System; cV5 Diagnostic Ultrasound System; V4 Diagnostic Ultrasound System; cV4 Diagnostic Ultrasound System	IYN
K251002	09/19/2025	VideaHealth Inc.	Videa Dental AI	MYN
K251071	05/02/2025	BrightHeart	Fetal EchoScan (v1.1)	POK
K251078	11/14/2025	Eos Imaging	AutoDensity	QIH
K251110	05/09/2025	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound Systems; Affiniti Series Diagnostic Ultrasound Systems	IYN
K251167	09/19/2025	Shanghai United Imaging Healthcare Co., Ltd.	uDR Aurora CX	KPR
K251195	01/27/2026	Aidoc Medical, Ltd.	BriefCase-Triage	QAS
K251276	05/21/2025	Hyperline, Inc.	Swoop® Portable MR Imaging® System	LNH
K251322	07/25/2025	GE Medical Systems Ultrasound and Primary care Diagnostics.	Venue; Venue Go; Venue Fit; Venue Sprint	IYN
K251368	09/12/2025	Diagnoly	FETOLY	IYN
K251370	12/01/2025	Canon Medical Systems Corporation	Cartesian Prime (PCD-1000A/3) V10.21	KPS
K251406	05/30/2025	Aidoc Medical, Ltd.	BriefCase-Triage	QAS
K251455	07/24/2025	Philips Ultrasound LLC	EPIQ Series Diagnostic Ultrasound System; Affiniti Series Diagnostic Ultrasound System	IYN
K251456	06/05/2025	BrightHeart	BrightHeart View Classifier	QIH
K251481	08/20/2025	Siemens Medical Solutions USA, Inc.	ACUSON Sequoia Diagnostic Ultrasound System; ACUSON Sequoia Select Diagnostic Ultrasound System; ACUSON Origin Diagnostic Ultrasound System; ACUSON Origin ICE Diagnostic Ultrasound System	IYN

FDA pediatric-use inventory: label-positive candidates (181–200 of 230)

Submission	Date	Company	Device	Code
K251514	12/05/2025	Overjet, Inc.	Overjet CBCT Assist	QIH
K251532	11/03/2025	Mighty Oak Medical	Acorn 3D Software (AC-SEG-4009); Acorn 3DP Model (AC-101-XX)	LLZ
K251651	10/09/2025	Philips Ultrasound, LLC	EPIQ Series Diagnostic Ultrasound System; Affiniti Series Diagnostic Ultrasound System	IYN
K251673	10/17/2025	X9, Inc.	X9 Ultrasound System	SGH
K251682	09/09/2025	Springbok, Inc. (dba Springbok Analytics)	MuscleView 2.0	LNH
K251747	08/15/2025	EOS Imaging	VEA Align; spineEOS	QIH
K251963	10/29/2025	Ge Medical Systems Ultrasound and Primary Care Diagnostics	LOGIQ E10s	IYN
K251985	10/29/2025	Ge Medical Systems Ultrasound and Primary Care Diagnostics	LOGIQ E10	IYN
K252103	12/02/2025	Mighty Oak Medical	Acorn 3D Software (AC-SEG-4009); Acorn 3DP Model (AC-101-XX)	QIH
K252148	03/27/2026	Butterfly Network, Inc.	Butterfly Gestational Age Tool	IYN
K252188	01/15/2026	Elekta Solutions AB	EMLA (Elekta Evo); EMLA (VersaHD); EMLA (Elekta Harmony Pro); EMLA (Elekta Infinity); EMLA (Elekta Harmony); EMLA (Elekta Synergy)	IYE
K252237	04/17/2026	Edgecare, Inc.	EdgeFlow UW20	IYO
K252294	12/08/2025	Brightheart	Fetal EchoScan (v1.2)	POK
K252328	11/24/2025	Ge Medical Systems Ultrasound and Primary Care Diagnostics	Voluson Expert 18; Voluson Expert 20; Voluson Expert 22	IYN
K252362	08/22/2025	Galileo CDS, Inc	GBrain MRI	QIH
K252433	03/16/2026	Sonio	Sonio Detect (v3)	IYN
K252503	04/30/2026	Canon, Inc.	Intelligent NR	OWB
K252538	03/05/2026	Orca Dental AI, Ltd.	CEPHX3D	QIH
K252539	09/03/2025	Arterys, Inc.	Tempus Pixel	QIH
K252557	12/22/2025	Philips Ultrasound, LLC	Lumify Diagnostic Ultrasound System	IYN

FDA pediatric-use inventory: label-positive candidates (201–220 of 230)

Submission	Date	Company	Device	Code
K252558	05/05/2026	Philips Ultrasound, LLC	Lumify Diagnostic Ultrasound System	IYN
K252954	05/04/2026	Neurocareai, Inc. (Dba Savellife.Ai)	MammoSightAI	QFM
K252970	01/07/2026	Aidoc Medical, Ltd.	BriefCase-Triage: CARE Multi-triage CT Body	QAS
K253265	11/06/2025	Aidoc Medical, Ltd.	BriefCase-Triage	QAS
K253269	11/26/2025	Ge Hualun Medical Systems Co., Ltd.	OEC One CFD	OXO
K253366	01/07/2026	GE Medical Systems Ultrasound and Primary Care Diagnostics	LOGIQ Fortis	IYN
K253370	01/08/2026	GE Medical Systems Ultrasound and Primary Care Diagnostics	LOGIQ Totus	IYN
K253489	12/12/2025	Hyperfine, Inc.	Swoop® Portable MR Imaging® System	LNH
K253502	04/14/2026	Ge Medical Systems, LLC	Critical Care Suite with Enteric Tube Positioning AI Algorithm	QIH
K253578	02/26/2026	Aidoc Medical, Ltd.	BriefCase-Triage: CARE Multi-Triage CT for Pneumothorax; Pericardial effusion; Large aortic aneurysm; Shoulder fracture or dislocation device	QAS
K253595	03/27/2026	Philips Ultrasound, LLC	EPIQ Series Diagnostic Ultrasound System, Affiniti Series Diagnostic Ultrasound System	IYN
K253649	03/27/2026	Philips Medical Systems Technologies, Ltd.	Spectral CT Verida Family	JAK
K253689	04/10/2026	Siemens Healthcare GmbH	syngo Dynamics (VA41F)	QIH
K253818	03/03/2026	Harrison-AI Medical Pty, Ltd.	Annalise Enterprise	QAS
K254015	04/01/2026	Smart Soft Healthcare AD	CoLumbo C-Spine	QIH
K260123	05/27/2026	Philips Ultrasound, LLC	EPIQ Series Diagnostic Ultrasound System	IYN
K260300	06/15/2026	Anumana, Inc.	WatchMate Software	QIH
K260322	06/11/2026	Mighty Oak Medical	Acorn 3D Software (AC-SEG-4009); Acorn 3DP Model (AC-101-XX)	QIH
K260378	05/12/2026	AZmed	Rayvolve	QBS
K260497	06/04/2026	Keya Medical Technology Co., Ltd.	DEEPPVESSEL Plaque	QIH

FDA pediatric-use inventory: label-positive candidates (221–230 of 230)

Submission ID	Date	Company	Device	Code
K260509	03/19/2026	Radformation, Inc.	AutoContour (RADAC V5)	QKB
K260667	06/29/2026	Philips Ultrasound, LLC	Alturion Series Diagnostic Ultrasound System	IYN
K260673	03/24/2026	GE Medical Systems Ultrasound and Primary Care Diagnostics	LOGIQ Vita; LOGIQ Vita Pro; LOGIQ Vita Express; LOGIQ Vita Plus; LOGIQ Vita Power; LOGIQ S20; LOGIQ S20 Pro; LOGIQ S20 Express; LOGIQ S20 Plus; LOGIQ S20 Power	IYN
K260680	05/27/2026	Philips Ultrasound, LLC	EPIQ Series Diagnostic Ultrasound Systems; Affiniti Series Diagnostic Ultrasound Systems	IYN
K260729	06/18/2026	GuideAI Health	Vascular Assist Occlusion Triage (VAOT) (v1.0)	QAS
K260844	06/16/2026	Siemens Medical Solutions USA, Inc.	ACUSON Sequoia Diagnostic Ultrasound System; ACUSON Sequoia Select Diagnostic Ultrasound System; ACUSON Origin Diagnostic Ultrasound System; ACUSON Origin ICE Diagnostic Ultrasound System	IYN
K261273	05/15/2026	Neurophat, Inc.	Neurophat AQUA	QIH
K261317	05/14/2026	Doc Medical, Ltd.	BriefCase-Triage	QAS
K261519	06/04/2026	Sonio	Sonio Suspect	POK
P150043	11/09/2018	View Medical, Inc.	QVCAD System	MYN

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Submission	Date	Company	Device	Code
DEN170022	07/19/2017	Quantitative Insights, Inc.	QuantX	POK
DEN190040	02/07/2020	Bay Labs, Inc.	Caption Guidance	QJU
DEN220040	01/12/2024	Imvria, Inc.	Fibresolve	QWO
DEN230023	04/09/2024	16 Bit Inc	Rho	SAO
DEN240047	05/30/2025	Clairity, Inc.	Allix5	SEZ
K111776	12/28/2011	RIVERAIN MEDICAL GROUP,LLC	DELTAVIEW MODEL 2.1	LLZ
K129526	12/27/2012	RIVERAIN TECH-NOLOGIES	CLEARREAD +CONFIRM	LLZ
K163138	11/30/2016	Clarius Mobile Health Corp.	Clarius Ultrasound System	IYN
K180589	04/05/2018	Agfa HealthCare N.V.	DR 800 with MUSICA Dynamic	JAA
K180599	05/02/2018	GE Medical Systems Ultrasound and Primary Care Diagnostics.	Venue	IYN
K193170	12/13/2019	GE Healthcare Japan Corporation	Deep Learning Image Reconstruction	JAK
K200497	07/16/2020	GE Medical Systems Ultrasound and Primary Care Diagnostics	Vivid S60N, Vivid S70N	IYN
K200708	09/09/2020	GE Medical Systems Ultrasound and	Vivid iq	IYN
K200743	07/23/2020	GE Medical Systems Ultrasound and	Vivid E80/ Vivid E90/ Vivid E95	IYN
K200851	09/09/2020	GE Medical Systems Ultrasound and Primary Care Diagnostics	Vivid T8, Vivid T9	IYN
K200852	09/18/2020	GE Medical Systems Ultrasound and Primary Care Diagnostics	EchoPAC Software Only, EchoPAC Plug-In	LLZ
K210438	09/10/2021	GE Medical Systems Ultrasound and Primary Care Diagnostics.	Versana Premier	IYN
K212067	09/17/2021	GE Healthcare Japan Corporation	Deep Learning Image Reconstruction	JAK
K212333	01/24/2022	Canon Medical Systems Corporation	Apilo i900/i800/i700 Diagnostic Ultrasound System, Software V6.5	IYN
K212690	12/21/2021	Qure.ai Technologies	qXR-BT	QIH

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Submission	Date	Company	Device	Code
K212960	03/22/2022	Canon Medical Systems Corporation	Apilo a550, Apilo a450, and Apilo a, Diagnostic Ultrasound System, Software V6.5	IYN
K213642	01/13/2022	GE Healthcare	Voluson S6, Voluson S6, Voluson S8t, Voluson S10, Voluson S10 Expert	IYN
K213999	02/18/2022	GE Medical Systems, LLC	Deep Learning Image Reconstruction	JAK
K220446	05/11/2022	GE Medical Systems Ultrasound and Primary Care Diagnostics	Versana Balance	IYN
K220619	07/15/2022	GE Medical Systems Ultrasound and Primary Care Diagnostics	Vivid S60N, Vivid S70N	IYN
K220862	07/22/2022	GE Medical Systems Ultrasound and	Vivid E80, Vivid E90, Vivid E95	IYN
K220940	07/22/2022	GE Medical Systems Ultrasound and Primary Care Diagnostics	EchoPAC Software Only, EchoPAC Plug-in	GIH
K221147	07/18/2022	GE Medical Systems Ultrasound and Primary Care Diagnostics	Vivid T8, Vivid T9	IYN
K221148	07/18/2022	GE Medical Systems Ultrasound & Primary Care Diagnostics LLC	Vivid iQ	IYN
K230807	04/20/2023	GE Healthcare Japan Corporation	Deep Learning Image Reconstruction	JAK
K231989	11/07/2023	GE Medical Systems Ultrasound and Primary Care Diagnostic	LOGIQ E10x, LOGIQ Fortis	IYN
K232361	12/07/2023	GE Medical Systems Ultrasound and Primary Care Diagnostics	LOGIQ Totus	IYN
K233195	01/24/2024	Canon Medical Systems Corporation	Apilo 900, Apilo 850 and Apilo 1700 Software V8.1 Diagnostic Ultrasound System	IYN
K233692	03/07/2024	GE Medical Systems Ultrasound and Primary Care Diagnostics	Voluson Signature 20, Voluson Signature 18	IYN
K241562	09/12/2024	Canon Medical Systems Corporation	Apilo 900/850/1700 Diagnostic Ultrasound System, Software V7.0 (TUS-A900, TUS-A800, TUS-A700)	IYN
K241593	02/05/2025	Gleamer SAS	BoneMetrics (US)	GIH
K241671	05/16/2025	Esate S.p.A.	6450 Ultrasound System (MyLabE80); 6450 Ultrasound System (MyLabE85)	IYN
K242005	10/02/2024	GE Medical Systems Ultrasound and Primary Care Diagnostics	Versana Premier, Versana Premier Lotus, LOGIQ F	IYN
K243620	02/11/2025	GE Medical Systems Ultrasound and Primary Care Diagnostics	Vivid iQ	IYN
K243628	02/11/2025	GE Medical Systems Ultrasound and Primary Care Diagnostics	Vivid T9/Vivid T8	IYN

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Submission Number	Date	Company	Device	Code
K250330	11/03/2025	Pie Medical Imaging BV	3mensio Workstation	QIH
K250543	05/29/2025	GE Medical Systems Ultrasound and Primary Care Diagnostics	Voluson Performance 16; Voluson Performance 18	IYN
K251106	08/29/2025	Fujifilm Sonosite Inc	Sonosite LX and Sonosite PX Ultrasound Systems	IYN
K251169	07/10/2025	GE Medical Systems Ultrasound and Primary Care Diagnostics	Vivid Pioneer	IYN
K251342	07/16/2025	GE Medical Systems Ultrasound and Primary Care Diagnostics	EchoPAC Software Only / EchoPAC Plug-in	QIH
K252074	10/31/2025	Canon Medical Systems Corporation	Aplo i900, Aplo i800 and Aplo i700 Software V9.0 Diagnostic Ultrasound System	IYN
K253050	06/18/2025	Informal	RadOncAI	MUJ
K253288	04/06/2025	Esate, S.p.A.	6450 Ultrasound System (MyLabE80); 6450 Ultrasound System (MyLabE85)	IYN
K253720	05/21/2025	Wuhan United Imaging Healthcare Co., Ltd	uSONIQUE Nova, uSONIQUE Nova Elite, uSONIQUE Vita, uSONIQUE Vita Elite, uSONIQUE Grace, uSONIQUE Grace Elite	IYN
K254013	05/14/2025	Subtle Medical, Inc.	SubtleHD-PET (1.x)	LLZ
K254120	05/15/2025	Subtle Medical, Inc.	SubtleHD-CT (1.x)	QIH
P160009	03/24/2025	TCAD Inc	PowerLook® Tomo Detection Software	MYN
P200003	01/11/2025	Seno Medical Instruments, Inc.	Imagio Breast Imaging System	QNK

Worth knowing as a radiologist — and why

- **Pediatric CT reconstruction (Brady et al., Radiology 2021)** (paper): a concrete dose-reduction application: lower dose with improved image quality in the studied protocols
- **RSNA Pediatric Bone Age Challenge (2017)** (benchmark / dataset): a landmark pediatric benchmark; useful for understanding automated bone-age accuracy and its limits
- **Gleamer BoneView / AZmed Rayvolve** (clinical products): examples of fracture-detection assistance with pediatric indications; eligibility depends on age, anatomy, and product version
- **FDA AI-enabled device list and linked labeling** (regulatory resource): a starting point for checking authorized use and pediatric age ranges; the list is not exhaustive
- **CLAIM checklist (2024 update)** (reporting guideline): check whether data, reference standards, and testing are adequately described; reporting completeness alone does not establish clinical benefit
- **Cross-hospital pneumonia testing (Zech et al., PLoS Med 2018)** (paper): models can exploit hospital-specific cues; ask for external testing and performance in your own patient population
- **TotalSegmentator** (research tool): automated CT / MR organ segmentation for volumes and body composition; check accuracy in children before trusting measurements
- **nnU-Net** (research framework): a strong baseline for training segmentation models; ask whether a new method improves on an appropriate comparator
- **MedSAM / segment-anything models** (research models): prompt-guided contours can assist annotation; radiologists still need to review and correct the output
- **MONAI** (research framework): infrastructure for imaging-AI development; useful when collaborating with a research team, rather than a ready-to-use diagnostic tool

Recommendations for pediatric radiologists

- Choose one task worth improving in your practice: bone-age measurements, fracture detection, or reducing CT dose / MRI scan time.
- Before adopting a tool, ask which children it was tested on. Check the intended age range and test it on your own cases, including younger children and unusual anatomy.
- In a local pilot, measure what changes with AI: reading time, missed findings, false alarms, dose, or scan time. Review errors and keep checking after rollout.
- Help build the evidence we need: choose a pediatric problem you see often, label cases carefully, and work with other hospitals to test whether a model works beyond the site that built it.